Now we want to show that $g(z)$ is analytic in the upper half-plane. Using

$$g(z) = \frac{1}{\sqrt{2\pi}} \int_0^\infty G(t) e^{izt}\, dt = \frac{1}{\sqrt{2\pi}} \int_0^\infty G(t) e^{i\omega t} e^{-\omega' t}\, dt\,, \tag{6.26}$$

we see that for $\omega' > 0$,

$$\frac{d^n g}{dz^n} = \frac{1}{\sqrt{2\pi}} \int_0^\infty G(t) \frac{d^n}{dz^n} e^{izt}\, dt = \frac{i^n}{\sqrt{2\pi}} \int_0^\infty t^n G(t) e^{i\omega t} e^{-\omega' t}\, dt\,, \tag{6.27}$$

since in this case the integrals in both (6.26) and (6.27) are uniformly convergent because of the term $e^{-\omega' t}(\omega' > 0, t > 0)$. Thus $g(z)$ is analytic in the upper half-plane $(\omega' > 0)$. However, it is clear that our assumptions on $G(t)$ do not enable us to extend the domain of analyticity to $\omega' \geq 0$. Nevertheless, we can say that $g(z)$ is bounded on the real axis, so the only singularities when $\omega' = 0$ will be of the branch point variety, and even then such branch singularities as $1/\sqrt{z}$ or $\log z$ are excluded by the boundedness requirement. The reader can see by looking back at the derivation of the original pair of Hilbert transform equations [Eqs. (6.19a) and (6.19b)] that it can be modified to include bounded branch point singularities on the real axis by taking a small semicircular detour around any such point. Thus Eqs. (6.19a) and (6.19b) remain unaltered by the presence of such singularities (the branches can always be chosen to avoid the upper half-plane). To eliminate the possibility of these branch singularities, one would have to assume an exponential type falloff of $G(\tau)$ as $\tau \to \infty$. Hence for any $g(z)$ arising from a $G(t)$ which satisfies assumptions (a), (b), and (c), we may write

$$g_R(\omega) = \frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{g_I(\bar{\omega})}{\bar{\omega} - \omega}\, d\bar{\omega}\,, \tag{6.28a}$$

$$g_I(\omega) = -\frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{g_R(\bar{\omega})}{\bar{\omega} - \omega}\, d\bar{\omega}\,. \tag{6.28b}$$

Thus by making a few very reasonable assumptions about the system in question, we can show that the real and imaginary parts of the physical quantity $g(\omega)$ are intimately related to each other for *real* values of the argument by what is essentially a dispersion relation. The key assumption is the *causality requirement*; we may say that causality implies the existence of dispersion relations in the case we have considered. In actual practice, one often restricts the term "dispersion relation" to mean an integral relation between two observable quantities which involves only an integration over values of the argument which are physically meaningful. Thus in Eqs. (6.28a) and (6.28b) only positive frequencies are accessible to experiment, so they are not directly useful as they stand. However, $G(t)$ is real, so we may proceed as follows:

$$g(z) = \frac{1}{\sqrt{2\pi}} \int_0^\infty G(t) e^{izt}\, dt\,,$$

$$g^*(z) = \frac{1}{\sqrt{2\pi}} \int_0^\infty G^*(t) e^{-iz^* t}\, dt = \frac{1}{\sqrt{2\pi}} \int_0^\infty G(t) e^{-iz^* t}\, dt = g(-z^*)\,.$$

Thus we have $g^*(z) = g(-z^*)$, which is often referred to as the *reality condition*. If z is real ($z = \omega$), we find that

$$g_R(\omega) - ig_I(\omega) = g_R(-\omega) + ig_I(-\omega) ,$$

or

$$g_R(\omega) = g_R(-\omega) , \tag{6.29a}$$

$$g_I(\omega) = -g_I(-\omega) , \tag{6.29b}$$

that is, g_R is an *even* function of ω and g_I is an *odd* function of ω. Note that if Eqs. (6.29a) and (6.29b) are satisfied, then the function

$$G(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(\omega) e^{-i\omega t} \, d\omega$$

is a real function.

Now in Eq. (6.28a), let us write

$$g_R(\omega) = \frac{1}{\pi} P \int_{-\infty}^{0} \frac{g_I(\bar{\omega})}{\bar{\omega} - \omega} \, d\bar{\omega} + \frac{1}{\pi} P \int_{0}^{\infty} \frac{g_I(\bar{\omega})}{\bar{\omega} - \omega} \, d\bar{\omega} .$$

In the first integral, we let $\bar{\omega} \to -\bar{\omega}$. Thus

$$g_R(\omega) = \frac{1}{\pi} P \int_{\infty}^{0} \frac{g_I(-\bar{\omega})}{\bar{\omega} + \omega} \, d\bar{\omega} + \frac{1}{\pi} P \int_{0}^{\infty} \frac{g_I(\bar{\omega})}{\bar{\omega} - \omega} \, d\bar{\omega} .$$

Using Eq. (6.29b), we finally obtain

$$g_R(\omega) = \frac{2}{\pi} P \int_{0}^{\infty} \frac{\bar{\omega} g_I(\bar{\omega})}{\bar{\omega}^2 - \omega^2} \, d\bar{\omega} , \tag{6.30a}$$

and in an identical manner,

$$g_I(\omega) = -\frac{2\omega}{\pi} P \int_{0}^{\infty} \frac{g_R(\bar{\omega})}{\bar{\omega}^2 - \omega^2} \, d\bar{\omega} . \tag{6.30b}$$

These expressions involve only positive, experimentally accessible frequencies. For the electric susceptibility, for example, we have

$$\chi_R(\omega) = \frac{2}{\pi} P \int_{0}^{\infty} \frac{\bar{\omega} \chi_I(\bar{\omega})}{\bar{\omega}^2 - \omega^2} \, d\bar{\omega} , \tag{6.31a}$$

$$\chi_I(\omega) = -\frac{2\omega}{\pi} P \int_{0}^{\infty} \frac{\chi_R(\bar{\omega})}{\bar{\omega}^2 - \omega^2} \, d\bar{\omega} . \tag{6.31b}$$

Equations (6.31a) and (6.31b) were first derived by H. A. Kramers and R. de L. Kronig and are referred to as the Kramers-Kronig dispersion relations.

Now, according to electromagnetic theory, we may write

$$n^2(\omega) = 1 + 4\pi\chi(\omega) ,$$

where $n(\omega)$ is the (complex) index of refraction. Since $\chi(z)$ is analytic in the upper half-z-plane, so is $n^2(z)$, and the function

$$n(z) = \sqrt{1 + 4\pi\chi(z)} \tag{6.32}$$

is also analytic in this region *if* $1 + 4\pi\chi(z)$ has no zeroes in the upper half-plane. If $1 + 4\pi\chi(z)$ vanishes for some $z = z_0$ in the upper half-plane, then according to the reality condition

$$\chi^*(z_0) = \chi(-z_0^*) ,$$

we find that $1 + 4\pi\chi(z)$ also vanishes at $z = -z_0^*$. Since $-z_0^*$ lies in the upper half-plane with the imaginary part equal to Im z_0, there will be a cut in the upper half-plane running from $-z_0^*$ to z_0. In this case, the dispersion relations of Eqs. (6.31a) and (6.31b) would have to be modified. We will assume that $n(z)$ has no zeroes in the upper half-plane and is therefore analytic in this region. However, because of Eq. (6.32), $|n(z)|$ does not tend to zero as $|z| \to \infty$. In fact,

$$n(z) \to 1 \qquad \text{as} \qquad |z| \to \infty$$

in the upper half-plane. This necessitates a modification in the treatment of the term coming from the large semicircle of Fig. 6.12 in obtaining Eq. (6.18). We have, in the case of $n(z)$,

$$\int_{S_R} \frac{n(z)}{z - \alpha}\, dz \xrightarrow{|z|\to\infty} \int_0^\pi \frac{dz}{z} = i\pi$$

in the notation of the previous section (see Fig. 6.12). Thus, for the case of $n(z)$, Eq. (6.18) must be replaced by

$$i\pi n(\omega) = i\pi + P\int_{-\infty}^{\infty} \frac{n(\bar{\omega})}{\bar{\omega} - \omega}\, d\bar{\omega} .$$

Separating real and imaginary parts, we obtain

$$n_R(\omega) = 1 + \frac{1}{\pi} P\int_{-\infty}^{\infty} \frac{n_I(\bar{\omega})}{\bar{\omega} - \omega}\, d\bar{\omega} , \tag{6.33a}$$

$$n_I(\omega) = -\frac{1}{\pi} P\int_{-\infty}^{\infty} \frac{n_R(\bar{\omega})}{\bar{\omega} - \omega}\, d\bar{\omega} . \tag{6.33b}$$

Making use of the reality condition, we can write these equations as

$$n_R(\omega) = 1 + \frac{2}{\pi} P\int_0^{\infty} \frac{\bar{\omega} n_I(\bar{\omega})}{\bar{\omega}^2 - \omega^2}\, d\bar{\omega} , \tag{6.34a}$$

$$n_I(\omega) = -\frac{2\omega}{\pi} P\int_0^{\infty} \frac{n_R(\bar{\omega})}{\bar{\omega}^2 - \omega^2}\, d\bar{\omega} . \tag{6.34b}$$

The quantity $\mu(\omega) \equiv (2\omega/c)n_I(\omega)$, when c is the speed of light in vacuum, is called the *absorption coefficient* and is the inverse of the distance a wave

$$\psi(x, t) = A \exp\left\{i\omega\left[\frac{n(\omega)}{c}x - t\right]\right\}$$

travels before its intensity drops to $1/e$ of its value at $x = 0$. In terms of μ, we have

$$n_R(\omega) = 1 + \frac{c}{\pi} P\int_0^{\infty} \frac{\mu(\bar{\omega})}{\bar{\omega}^2 - \omega^2}\, d\bar{\omega} . \tag{6.35}$$

Thus the real part of the index of refraction is completely specified by knowing the absorption coefficient at all frequencies ! It is from Eq. (6.35) that dispersion relations derive their name. Equation (6.35) relates a substance's absorption to its dispersive effects, i.e., to the way the real index of refraction varies with frequency. It is this variation with frequency which produces the well-known separation (dispersion) of different wavelengths of light by a prism.

In more recent times, the term dispersion relation has continued to be used to denote any relationship between real and imaginary parts of a physical quantity (a scattering amplitude in quantum mechanics, for example) which has the general appearance of a Hilbert transform. Note that since $\mu(\omega)$ must be positive for all frequencies on physical grounds (i.e., we do not expect to find waves which grow in time as they pass through a substance), Eq. (6.35) specifies that

$$n_R(0) = 1 + \frac{c}{\pi} P \int_0^\infty \frac{\mu(\bar{\omega})}{\bar{\omega}^2} d\bar{\omega} , \tag{6.36}$$

so we see that $n_R(0) > 1$. Thus the familiar static dielectric constant ϵ, given by $\epsilon = n^2(0)$ is always greater than unity. Since $\chi_I(0) = 0$, we see that the integral in Eq. (6.36) converges without our using the principal-value technique. Thus using $n(0) = \sqrt{\epsilon}$, we may write Eq. (6.36) as

$$\sqrt{\epsilon} - 1 = \frac{c}{\pi} \int_0^\infty \frac{\mu(\bar{\omega})}{\bar{\omega}^2} d\bar{\omega} .$$

This relation is known as a "sum rule" for the absorption coefficient; it relates the weighted integral over all values of the absorption coefficient to a simple, experimentally accessible constant.

The derivation of the dispersion relation for the index of refraction exhibits certain features which are often encountered in deriving dispersion relations. Namely, it often happens that the quantity in question does not tend toward zero as $|z|$ tends toward infinity, and, furthermore, one is not usually fortunate enough to know the precise behavior of the quantity as $|z|$ tends to infinity, except, for example, to say that it is *bounded* for large values of $|z|$. In this case, we can proceed as follows. Let α_0 be some point on the real axis at which $f(z)$ is analytic. Then the function

$$\frac{f(z) - f(\alpha_0)}{z - \alpha_0} \equiv \phi(z)$$

is not singular at $z = \alpha_0$, and $|\phi(z)| \to 0$ as $|z| \to \infty$. Also, if $f(z)$ is analytic in the upper half-plane, so is $\phi(z)$, and we can write a dispersion relation for $\phi(z)$:

$$i\pi \left[\frac{f(\alpha) - f(\alpha_0)}{\alpha - \alpha_0} \right] = P \int_{-\infty}^{\infty} \frac{f(x) - f(\alpha_0)}{(x - \alpha)(x - \alpha_0)} dx .$$

But

$$\frac{1}{(x - \alpha)(x - \alpha_0)} = \frac{1}{\alpha - \alpha_0} \left[\frac{1}{x - \alpha} - \frac{1}{x - \alpha_0} \right] .$$

Therefore, our dispersion relation takes the form

$$i\pi f(\alpha) = i\pi f(\alpha_0) + (\alpha - \alpha_0) P \int_{-\infty}^{\infty} \frac{f(x)\, dx}{(x - \alpha)(x - \alpha_0)} - f(\alpha_0) P \int_{-\infty}^{\infty} \frac{dx}{x - \alpha}$$
$$+ f(\alpha_0) P \int_{-\infty}^{\infty} \frac{dx}{x - \alpha_0} .$$

According to the work of the previous section, these last two principal-value integrals vanish, so we have just

$$i\pi f(\alpha) = i\pi f(\alpha_0) + (\alpha - \alpha_0) P \int_{-\infty}^{\infty} \frac{f(x)\, dx}{(x - \alpha)(x - \alpha_0)} .$$

Separating the real and imaginary parts, we finally obtain

$$f_R(\alpha) = f_R(\alpha_0) + \frac{1}{\pi} (\alpha - \alpha_0) P \int_{-\infty}^{\infty} \frac{f_I(x)}{(x - \alpha)(x - \alpha_0)} dx , \tag{6.37a}$$

$$f_I(\alpha) = f_I(\alpha_0) - \frac{1}{\pi} (\alpha - \alpha_0) P \int_{-\infty}^{\infty} \frac{f_R(x)}{(x - \alpha)(x - \alpha_0)} dx . \tag{6.37b}$$

Relations of the type of Eqs. (6.37a) and (6.37b) are referred to as *once-subtracted dispersion relations*. For them to be of use in a particular physical problem, one must have a means of determining, say, $f_R(\alpha_0)$ for *some* α_0 in addition to possessing the usual information required by an ordinary dispersion relation of the type of Eq. (6.19a). If the properties of $f(z)$ for large $|z|$ are even "worse" than assumed above [for example, suppose that $|f(z)/z|$ tended toward a nonzero constant as $|z|$ tended toward infinity], then one could introduce more subtraction points, α_1, α_2, etc, in a similar manner.

We have already seen that causality implies certain analyticity properties; we conclude this section by showing that the converse is also true, namely, analyticity implies causality! We will do this by using the analytic properties of $n(z)$ to show that electromagnetic signals will not propagate in any medium faster than the speed of light. Consider a wave front traveling in the x-direction in a dielectric medium with a complex index of refraction, $n(z)$. Assume that at $x = 0$ there is no disturbance before $t = 0$, that is, $\psi(0, t) = 0$ for $t < 0$. We can write a general wave, as $\psi(x, t)$, a superposition of plane waves of all frequencies:

$$\psi(x, t) = \int_{-\infty}^{\infty} \psi(\omega) \exp \left\{ i\omega \left[\frac{n(\omega)}{c} x - t \right] \right\} d\omega .$$

Note that

$$\psi(0, t) = \int_{-\infty}^{\infty} \psi(\omega) e^{-i\omega t}\, d\omega ,$$

so

$$\psi(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \psi(0, t) e^{i\omega t}\, dt . \tag{6.38}$$

We may now use Eq. (6.38) to define $\psi(z)$ for z complex:

$$\psi(z) \equiv \frac{1}{2\pi} \int_{0}^{\infty} \psi(0, t) e^{i\omega t} e^{-\omega' t}\, dt ,$$

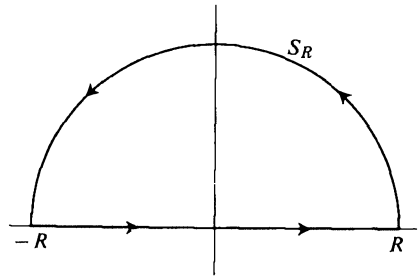

Fig. 6.13 The contour, C, used in evaluating Eq. (6.39). R may become as large as we please.

where we have made use of the fact that $\psi(0, t) = 0$ for $t < 0$ and have written, as usual, $z = \omega + i\omega'$. By our previous arguments, $\psi(z)$ is analytic in the upper half-plane and tends in norm toward zero as $|z|$ tends toward infinity in this region. Now let us consider

$$\oint_C \psi(z) \exp\left[iz\left(\frac{n(z)}{c}x - t\right)\right] dz$$

around the contour shown in Fig. 6.13. Since $\psi(z)$ and $n(z)$ are analytic in the upper half-plane,

$$\oint_C \psi(z) \exp\left[iz\left(\frac{n(z)}{c}x - t\right)\right] dz = 0 .$$

Thus

$$\int_{-R}^{R} \psi(\omega) \exp\left[i\omega\left(\frac{n(\omega)}{c}x - t\right)\right] d\omega + \int_{S_R} \psi(z) \exp\left[iz\left(\frac{n(z)}{c}x - t\right)\right] dz = 0 . \tag{6.39}$$

Calling the second integral in Eq. (6.39) I_R, we have

$$|I_R| \leq \int_0^{\pi} |\psi(z)| \exp\left[-R\sin\theta\left(\frac{n_R}{c}x - t\right) - R\cos\theta\,\frac{n_I}{c}x\right] R\,d\theta ,$$

where R is some arbitrarily large number. Now as $R \to \infty$, $n_I \to 0$ and $n_R \to 1$, as we have seen above. Thus, under these circumstances,

$$|I_R| \leq 2R \int_0^{\pi/2} |\psi(z)| \exp\left[-R\left(\frac{x}{c} - t\right)\sin\theta\right] d\theta .$$

But $\sin\theta \geq 2\theta/\pi$ for $0 \leq \theta \leq \pi/2$. (To see this, note that $g(\theta) = \sin\theta - 2\theta/\pi$ vanishes at $\theta = 0$ and $\theta = \pi/2$. It is positive at $\theta = \pi/4$, so if it is to become zero or negative inside $[0, \pi/2]$, it must take on a minimum in this region. However, $g''(\theta)$ is always *negative* in $[0, \pi/2]$ so the function cannot possibly take on a minimum value. This can also be seen by drawing a graph of $\sin\theta$ and $2\theta/\pi$.) Therefore, if $(x/c - t)$ is positive,

$$|I_R| \leq 2R \int_0^{\pi/2} |\psi(z)| \exp\left[-\frac{2R}{\pi}\left(\frac{x}{c} - t\right)\theta\right] d\theta .$$

Now $|\psi(z)| \to 0$ as $R \to \infty$, so let us write for large R, $|\psi(z)| \leq aR^{-\lambda}(\lambda > 0)$. Then

$$|I_R| \leq 2aR^{1-\lambda} \int_0^{\pi/2} \exp\left[-\frac{2}{\pi} R\left(\frac{x}{c} - t\right)\theta\right] d\theta ,$$

or

$$|I_R| \leq \frac{\pi a}{R^{\lambda}[(x/c) - t]}\left[1 - \exp\left\{-R\left(\frac{x}{c} - t\right)\right\}\right].$$

If we assume that $x/c > t$, then as $R \to \infty$, we see that $|I_R| \to 0$. Note that if $x/c < t$, we cannot draw this conclusion because of the *growing* exponential in the above inequality. Thus as $R \to \infty$, Eq. (6.39) becomes

$$\psi(x, t) = \int_{-\infty}^{\infty} \psi(\omega) \exp\left\{i\omega\left[\frac{n(\omega)}{c} x - t\right]\right\} d\omega = 0$$

for $x/c > t$. Thus we reach the satisfying conclusion that if no signal is present at $x = 0$ when $t = 0$, then there will be no signal at $x = x_0 > 0$ before $t = x_0/c$, that is, a signal can propagate with at most the speed of light, c, even though $c/n(\omega)$ may be greater than c, since $n(\omega)$ is known experimentally to become less than 1 at high frequencies [it clearly *cannot* do so at very low frequencies, since we have already shown that $n(0) \geq 1$].

6.7 THE EXPANSION OF AN ANALYTIC FUNCTION IN A POWER SERIES

We now come to one of the most important applications of the Cauchy-Goursat theorem, namely, the possibility of expanding an analytic function in a power series. The main result may be stated as follows:

Laurent's theorem. Let $f(z)$ be analytic throughout the closed annular region between the two circles C_1 and C_2 with common center z_0. Then at each point in this annulus

$$f(z) = \sum_{n=-\infty}^{\infty} A_n (z - z_0)^n , \tag{6.40}$$

with the series converging uniformly in any closed region, R, lying wholly within the annulus. Here

$$A_n = \frac{1}{2\pi i} \oint_C \frac{f(z')}{(z' - z_0)^{n+1}} dz' , \tag{6.41}$$

for $n = 0, \pm 1, \pm 2, \cdots$, and C is any closed contour in the annulus which encircles z_0.

Proof. Consider the contour K enclosing the region R as shown in Fig. (6.14); it may be written symbolically as $K = C_1 + L_1 - C_2 + L_2$. Here we adhere to the convention that simple circular contours are always traversed in a counterclockwise direction. Since the inner circle is traversed in a clockwise direction

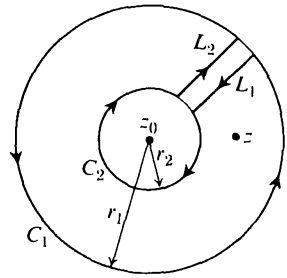

Fig. 6.14

when considered as a part of K, we write it is $-C_2$, where C_2 is a conventionally oriented curve. Since $f(z)$ is analytic within and on K,

$$2\pi i f(z) = \oint_K \frac{f(z')}{z' - z} dz' = \int_{C_1} \frac{f(z')}{z' - z} dz'$$
$$+ \int_{L_1} \frac{f(z')}{z' - z} dz + \int_{-C_2} \frac{f(z')}{z' - z} dz' + \int_{L_2} \frac{f(z')}{z' - z} dz' ,$$

where $z \in R$. By the same argument used in deriving Cauchy's integral formula we see that if L_1 and L_2 are taken to be arbitrarily close together the integrals along L_1 and L_2 cancel, and

$$f(z) = \frac{1}{2\pi i} \oint_{C_1} \frac{f(z')}{z' - z} dz' - \frac{1}{2\pi i} \oint_{C_2} \frac{f(z')}{z' - z} dz' , \tag{6.42}$$

where we have used Eq. (6.11). Equation (6.42) is the starting point for the proof of Laurent's theorem. To proceed further we make use of the identity (for $\alpha \neq 0$)

$$\frac{1}{\alpha - \beta} = \frac{1}{\alpha} + \frac{\beta}{\alpha(\alpha - \beta)} .$$

The second term is just $(\beta/\alpha)[1/(\alpha - \beta)]$, that is, β/α times the originally expanded quantity, so upon iterating N times we get

$$\frac{1}{\alpha - \beta} = \sum_{n=0}^{N} \frac{\beta^n}{\alpha^{n+1}} + \left(\frac{\beta}{\alpha}\right)^{N+1} \frac{1}{\alpha - \beta} . \tag{6.43}$$

Keeping this last equation in mind, let us write Eq. (6.42) in the form

$$f(z) = \frac{1}{2\pi i} \oint_{C_1} \frac{f(z')}{[(z' - z_0) - (z - z_0)]} dz' + \frac{1}{2\pi i} \oint_{C_2} \frac{f(z')}{[(z - z_0) - (z' - z_0)]} dz' .$$

Using Eq. (6.43) we can rewrite this as

$$f(z) = \frac{1}{2\pi i}\sum_{n=0}^{N}\oint_{C_1}\frac{(z-z_0)^n}{(z'-z_0)^{n+1}}f(z')\,dz' + \frac{1}{2\pi i}\oint_{C_1}\left[\frac{z-z_0}{z'-z_0}\right]^{N+1}\frac{f(z')}{z'-z}dz'$$
$$+ \frac{1}{2\pi i}\sum_{m=0}^{N}\oint_{C_2}\frac{(z'-z_0)^m}{(z-z_0)^{m+1}}f(z')\,dz' + \frac{1}{2\pi i}\oint_{C_2}\left[\frac{z'-z_0}{z-z_0}\right]^{N+1}\frac{f(z')}{z-z'}dz'. \tag{6.44}$$

According to by now familiar arguments the contours C_1 and C_2 in the first and third terms of the above equation may be replaced by any contour C about z_0 which is contained in the annulus bounded by C_1 and C_2. Making the change of index $m = n + 1$ in the third term, we obtain

$$f(z) = \sum_{n=-N-1}^{N} A_n(z-z_0)^n + R_N(z)\,,$$

where A_n is given by Eq. (6.41) and

$$R_N(z) = \frac{1}{2\pi i}\oint_{C_1}\left[\frac{z-z_0}{z'-z_0}\right]^{N+1}\frac{f(z')}{z'-z}dz' + \frac{1}{2\pi i}\oint_{C_2}\left[\frac{z'-z_0}{z-z_0}\right]^{N+1}\frac{f(z')}{z-z'}dz'.$$

If we can show that the magnitude of $R_N(z)$ can be made less than any pre-as signed ϵ for N sufficiently large, where N is independent of z, then the proof will be complete. Letting r_1 be the radius of C_1 and r_2 be the radius of C_2, we have

$$|R_N(z)| \le \frac{1}{2\pi}\left[\int_0^{2\pi}\frac{|z-z_0|^{N+1}}{r_1^{N+1}}\frac{|f(z')|}{|z'-z|}r_1\,d\theta + \int_0^{2\pi}\frac{r_2^{N+1}}{|z-z_0|^{N+1}}\frac{|f(z')|}{|z-z'|}r_2\,d\theta\right].$$

Now we define

$$M_1 \equiv \operatorname*{Max}_{z\in C_1}|f(z)|\,, \qquad M_2 \equiv \operatorname*{Max}_{z\in C_2}|f(z)|\,,$$
$$l_1 \equiv \operatorname*{Max}_{z\in R}|z-z_0|\,, \qquad l_2 \equiv \operatorname*{Min}_{z\in R}|z-z_0|\,,$$
$$d_1 \equiv \operatorname*{Min}_{\substack{z\in R\\ z'\in C_1}}|z'-z|\,, \qquad d_2 \equiv \operatorname*{Min}_{\substack{z\in R\\ z'\in C_2}}|z'-z|\,.$$

Since R is a domain within the annulus bounded by C_1 and C_2,

$$l_1 < r_1\,, \qquad l_2 > r_2\,, \tag{6.45}$$

Thus,

$$|R_N(z)| \le \frac{M_1 r_1}{d_1}\left(\frac{l_1}{r_1}\right)^{N+1} + \frac{M_2 r_2}{d_2}\left(\frac{r_2}{l_2}\right)^{N+1}. \tag{6.46}$$

But according to Eq. (6.45), $(l_1/r_1) < 1$ and $(r_2/l_2) < 1$, independent of z, so the magnitude of $R_N(z)$ can be made as small as we please for N sufficiently large. Since the bound of Eq. (6.46) is independent of z we have completed the proof

of the uniform convergence of the series in Eq. (6.40). This series is known as the *Laurent expansion* of $f(z)$ about the point z_0.

Example 1. Consider the function

$$f(z) = \frac{z^3 + 2z^2 + 4}{(z-1)^3}.$$

We shall obtain the Laurent expansion of $f(z)$ about the singular point $z = 1$. Writing $z \equiv (z-1) + 1$, we have

$$\begin{aligned} f(z) &= \frac{[(z-1)+1]^3 + 2[(z-1)+1]^2 + 4}{(z-1)^3} \\ &= 1 + \frac{5}{z-1} + \frac{7}{(z-1)^2} + \frac{7}{(z-1)^3}. \end{aligned}$$

Clearly this is the Laurent expansion of $f(z)$ about $z = 1$, and the reader is urged to verify this by evaluating the coefficients A_n of Eq. (6.41) which are non-vanishing for this particular $f(z)$.

Example 2. We now consider the Laurent series for a less trivial case, namely the function $\cosh(z + 1/z)$. The hyperbolic cosine is an entire function. Its argument, $z + 1/z$, is analytic everywhere except at the origin, and therefore $\cosh(z + 1/z)$ is analytic everywhere except at the origin. Thus we can pick C_2 to be an arbitrarily small circle about the origin and C_1 to be an arbitrarily large circle about the origin. Then

$$\cosh\left(z + \frac{1}{z}\right) = \sum_{-\infty}^{\infty} A_n z^n, \quad \text{where} \quad A_n = \frac{1}{2\pi i}\oint_C \frac{\cosh(z' + 1/z')}{(z')^{n+1}}\,dz',$$

and C is any closed contour about the origin. Let C be the unit circle. Then $z' = e^{i\theta}$ on C, so the integral becomes

$$\begin{aligned} A_n &= \frac{1}{2\pi}\int_{-\pi}^{\pi} \cosh(2\cos\theta)e^{-in\theta}\,d\theta \\ &= \frac{1}{\pi}\int_0^{\pi} \cosh(2\cos\theta)\cos n\theta\,d\theta. \end{aligned}$$

This integral may be evaluated by using the integral representation of the Bessel function, which will be discussed in Section 6.9. We will obtain A_n by still another method later in this section but will give the result here for the sake of completeness:

$$\begin{aligned} A_{2n} &= \sum_{m=0}^{\infty} \frac{1}{m!(m + 2|n|)!}, \\ A_{2n+1} &= 0 \quad \text{for} \quad n = 0, \pm 1, \pm 2, \cdots \end{aligned} \tag{6.47}$$

Example 3. To illustrate another kind of problem involving Laurent expansions, consider the function $f(z) = (z^2 - 1)^{-1/2}$. According to the discussion in

Section 6.2, this function has branch points at $z = -1$ and $z = +1$. We can choose the cut to run between these two points along the real axis, defining a single-valued branch by

$$f(z) = (\rho_1\rho_2)^{-1/2} e^{-i\phi_1/2} e^{-i\phi_2/2} ,$$

where $0 \leq \phi_1 < 2\pi$, $0 \leq \phi_2 < 2\pi$, $\rho_1 = |z - 1|$, $\rho_2 = |z + 1|$ and $(\rho_1\rho_2)^{-1/2}$ denotes the positive square root of $1/\rho_1\rho_2$ (see Fig. (6.15)). With these definitions $f(z)$ is analytic everywhere except on the real axis between $z = -1$ and $z = +1$. Thus, if we take C_2 to be a circle centered at $z = 0$ with radius infinitesimally larger than 1 and C_1 to be a circle of arbitrarily large radius, we can obtain a Laurent expansion of $f(z)$ in the annulus defined by C_1 and C_2. As usual

$$(z^2 - 1)^{-1/2} = \sum_{n=-\infty}^{\infty} A_n z^n ,$$

where

$$A_n = \frac{1}{2\pi i} \oint_C \frac{(z'^2 - 1)^{-1/2}}{(z')^{n+1}} dz' .$$

For $n \geq 0$ we choose C to be a circle of arbitrarily large radius; it is clear that for this contour the relevant integral vanishes, so $A_n = 0$ for $n \geq 0$. To deal with the case of negative n, we choose as contour any circle with radius greater than 1 (see Fig. 6.15). According to our results on path independence, this contour can be deformed into a "dogbone" contour as shown in Fig. 6.15. The

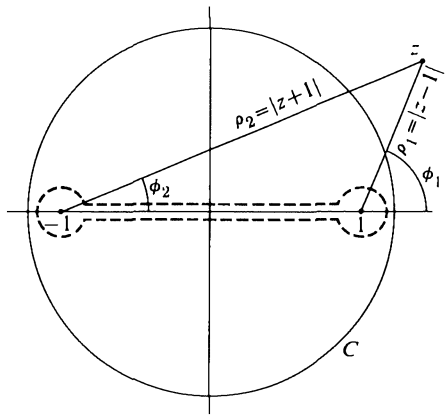

Fig. 6.15 The circular contour C is the starting point for the evaluation of the Laurent series coefficients, A_n of $(z^2 - 1)^{-1/2}$ when n is negative. ρ_1, ρ_2, ϕ_1, and ϕ_2 define the single-valued branch of this function, and the dashed "dogbone" contour is the deformation of C which enables us to evaluate the coefficients.

contribution to A_n from the infinitesimally small circles at the ends of the bone is vanishingly small, so we have simply

$$A_n = \frac{1}{2\pi i}\left[\int_1^{-1} \frac{1}{\sqrt{1-x^2}} e^{-i\pi/2} x^m \, dx + \int_{-1}^{1} \frac{1}{\sqrt{1-x^2}} e^{-i(\pi+2\pi)/2} x^m \, dx\right]$$
$$= \frac{1}{\pi}\int_{-1}^{1} \frac{x^m}{\sqrt{1-x^2}} dx ,$$

where $m = -n - 1$ (note that for the range of n in question m is never negative). Clearly, for m odd (n even) A_n vanishes. For m even (n odd) the integral is elementary; we have

$$A_n = \frac{2}{\pi}\int_0^1 \frac{x^m}{\sqrt{1-x^2}} dx = \frac{2}{\pi}\int_0^{\pi/2} \sin^m \theta \, d\theta = \frac{m!}{2^m[(m/2)!]^2} .$$

Thus,

$$(z^2-1)^{-1/2} = \sum_{m=0}^{\infty}{}' \frac{m!}{2^m[(m/2)!]^2} \frac{1}{z^{m+1}} ,$$

where the prime indicates that only even values of m are included in the summation. Setting $m = 2\nu$, we have finally

$$(z^2-1)^{-1/2} = \sum_{\nu=0}^{\infty} \frac{(2\nu)!}{4^\nu(\nu!)^2} \frac{1}{z^{2\nu+1}} = \frac{1}{z} + \frac{1}{2}\frac{1}{z^3} + \frac{3}{8}\frac{1}{z^5} + \frac{5}{16}\frac{1}{z^7} + \cdots .$$

This result is not surprising: it is just what we would have obtained by expanding $(z^2-1)^{-1/2}$ using real variable techniques. Note that in keeping with our definition of the single-valued branch of $(z^2-1)^{-1/2}$ the Laurent series gives a value of the square root which is positive on the positive real axis and negative on the negative real axis.

Until now we have spoken of singularities as points where the function in question is not analytic, but have made no attempt to classify singular points. There are differences, however, as we might expect if we look at the singular point $z = 0$ of the three functions $1/z$, $1/z^2$, and $\cosh(z + 1/z)$. We feel that the singularity of $1/z^2$ is a little worse than that of $1/z$ and that the singularity of $\cosh(z + 1/z)$ is terrible. A way of classifying the singularities is provided by the Laurent series of each function.

Suppose that $f(z)$ is analytic in a domain R except at $z = z_0$. Expand $f(z)$ in a Laurent series about z_0 inside R:

$$f(z) = \sum_{n=0}^{\infty} a_n(z-z_0)^n + \sum_{n=1}^{\infty} b_n \frac{1}{(z-z_0)^n} .$$

If $b_n = 0$ for $n = N+1, N+2, \cdots, \infty$, that is, if the series of negative powers of $(z - z_0)$ terminates with the Nth power, then $f(z)$ is said to have a *pole of order* N at z_0. The functions $1/z$ and $1/z^2$ have poles of order 1 and 2 respectively. But the Laurent expansion of $\cosh(z + 1/z)$ has an *infinite* number of negative powers of z; so does the Laurent expansion of $e^{1/z}$ as we shall soon

see. The singularities of these functions at the origin are called *essential* singularities. They cannot be removed by multiplying the function by some finite power of z, as is the case with the poles of $1/z$ and $1/z^2$. Notice, however, that the presence of an infinite number of negative powers of z in a Laurent series does not guarantee the presence of an essential singularity. Example 3 above shows that a branch cut singularity can also give rise to an infinite number of negative powers of z in a Laurent series. Only if we know that the singularity in question is confined within an arbitrarily small region can we conclude that an infinite number of negative powers of z implies the existence of an essential singularity. A function with no essential singularities in a region—although it may have poles there—is said to be a *meromorphic* function.

A particularily important consequence of Laurent's theorem arises when there are *no singularities* contained within the inner circle C_2, that is, if $f(z)$ is analytic everywhere inside C_1. Using Laurent's theorem, we can write

$$f(z) = \sum_{n=-\infty}^{\infty} A_n (z - z_0)^n,$$

where

$$A_n = \frac{1}{2\pi i} \oint_C \frac{f(z')}{(z' - z_0)^{n+1}} dz' . \tag{6.48}$$

But if $f(z')$ is analytic inside C_1, then for $n = -1, -2, \cdots$, the integrand in Eq. (6.48) is analytic within and on C (since C lies within C_1). Therefore, according to the Cauchy-Goursat theorem, $A_n = 0$ for $n \leqslant -1$. Furthermore, for $n \geqslant 0$, we have, using Eq. (6.17),

$$\frac{1}{2\pi i} \oint_C \frac{f(z')}{(z' - z_0)^{n+1}} dz' = \frac{1}{n!} f^{(n)}(z_0) ,$$

where $f^{(n)}(z_0)$ denotes the nth derivative of $f(z)$ at $z = z_0$ and $f^{(0)}(z) \equiv f(z_0)$. Thus,

$$f(z) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(z_0) (z - z_0)^n . \tag{6.49}$$

This result is known as Taylor's theorem, and the series in Eq. (6.49) is called Taylor's series for $f(z)$ about z_0. This very important result may be stated formally as follows:

Taylor's Theorem. If $f(z)$ is analytic at all points interior to a circle C centered about z_0 then in any closed region contained wholly inside C

$$f(z) = \sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(z_0) (z - z_0)^n , \tag{6.50}$$

and the series converges uniformly.

We can now express the analytic function $f(z)$ as a uniformly convergent series of analytic functions (it is a simple matter to show that *any* uniformly

convergent series of analytic functions is analytic). Because of the uniformity of the convergence, it can readily be shown that the integral of $f(z)$ along any path in the region of convergence of the power series in Eq. (6.49) can be evaluated by integrating the series term-by-term (see Problem 6.28). Using Eq. (6.17), we see that this also means that the derivatives of $f(z)$ can be evaluated similarly by term-by-term differentiation. Thus Eq. (6.49) provides $f(z)$ and all its derivatives throughout the region of convergence.

Examples

1. $e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \qquad \text{for} \quad |z| < \infty .$

2. $\sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!}, \qquad \text{for} \quad |z| < \infty .$

3. $(1-z)^{-1} = \sum_{n=0}^{\infty} z^n, \qquad \text{for} \quad |z| < 1 .$

These series all reduce to familiar results for real values of z.

It is often possible to obtain the *Laurent* expansion of some function by using a related Taylor series. Thus, Example 1, above, gives us

$$e^{1/z} = \sum_{n=0}^{\infty} \frac{1}{n!\, z^n} \qquad (|z| > 0) .$$

This show clearly that $e^{1/z}$ has an essential singularity at $z = 0$. A slightly more substantial example is provided by the Taylor expansion of $\cosh z$. It is a simple matter to show that

$$\cosh z = \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!}$$

For $|z| < \infty$. Thus for $|z| > 0$ we have

$$\cosh (z + 1/z) = \sum_{n=0}^{\infty} \frac{1}{(2n)!} (z + 1/z)^{2n} .$$

Using the binomial theorem, we find

$$\cosh (z + 1/z) = \sum_{n=0}^{\infty} \sum_{m=0}^{2n} \frac{z^{2(n-m)}}{(2n-m)!\, m!} = \sum_{n=0}^{\infty} \sum_{\mu=-n}^{n} \frac{z^{2\mu}}{(n+\mu)!\, (n-\mu)!} .$$

If in Eq. (6.50) we interchange the order of summation and then make the change of variable $\nu = n - \mu$, we obtain

$$\cosh (z + 1/z) = \sum_{\mu=-\infty}^{\infty} z^{2\mu} \left[\sum_{\nu=0}^{\infty} \frac{1}{(\nu + 2|\mu|)!\, \nu!} \right],$$

in agreement with the result of Eqs. (6.47).

The region in which a Taylor series expansion of a function about a point z_0 is valid is limited by the presence of a singularity of $f(z)$. We have seen that if this singularity is at the point α_0, the Taylor's expansion holds within a circular region of radius $|\alpha_0 - z_0|$ about z_0. Now the Taylor expansion gives the values of the analytic function and all its derivatives at every point in the region of analyticity. In particular, these quantities are known at a point z_1 (see Fig. 6.16) near the region's border. We may now use the point z_1 as a point about which to expand the function in another Taylor series. We can do this because $f^{(n)}(z_1)$ is known for all n from the first Taylor expansion about z_0. The radius of convergence of this second series expansion about z_1 is determined by the distance from z_1 to the nearest singularity. Continuing in this way (Fig. 6.16), we can determine the function throughout the entire plane except the points at which it is singular. To get started, we need only know the values of the analytic function in some *region*, however small. This process is called *analytic continuation*. It is as though a paleontologist could reconstruct a whole dinosaur from the fossil remains of a single toenail.

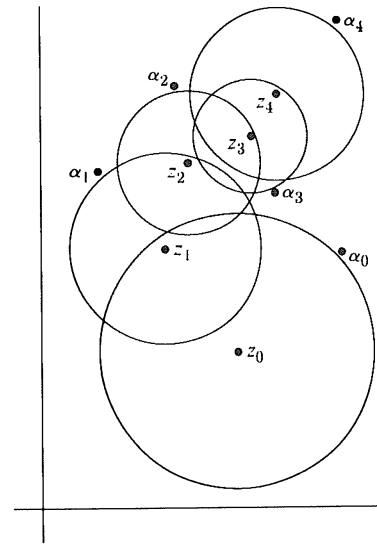

Fig. 6.16 A sequence of Taylor expansions which analytically continue a function originally known in some region around z_0. The first expansion about z_0 is limited in its radius of expansion by the singularity at α_0. The next Taylor expansion about z_1 (inside the first expansion's radius of convergence) is limited by a second singularity at α_1, and so on.

The process of analytic continuation is the best demonstration of the rigid inner structure of *analytic* functions. Again we see how interdependent are the values of an analytic function: its values in *any* region on the plane determine its values *everywhere* that it is analytic. This blueprint for the construction of all the values of an analytic function also demonstrates the necessity of the property that all derivatives of analytic functions are analytic. For if this were not the case, we could not continue a function analytically by a chain of Taylor expansions. To do this requires that we be able to approximate the function arbitrarily well, and this in turn means that all its derivatives must exist so the Taylor series can be extended to achieve arbitrary accuracy.

An immediate consequence of the fact that an analytic function may be continued is that a function which is zero along any curve in a region R, throughout which it is analytic, must be zero everywhere in R. This means that if two functions $f(z)$ and $g(z)$ are equal on any curve inside a simply-connected region R in which both are single-valued and analytic, then they are equal everywhere in R, since $f(z) - g(z)$ is zero on the curve and therefore throughout R. It will be recalled that the analytic functions e^z, $\sin z$, $\sinh z$, etc., were defined for complex arguments in a way that reduced to the usual definitions on the real axis. The principle discussed above shows that these functions could not have been defined otherwise away from the real axis and still be analytic. Furthermore, this explains why all the familiar identities satisfied by the functions for real values continue to hold throughout the complex plane.

It should also be noted that the process of analytic continuation is closely related to the problem of multivalued functions and Riemann surfaces discussed in Section 6.2. Suppose that we are given an analytic power series for a function in some region, for example,

$$w(z) = \sqrt{z} = \sqrt{1 - (1 - z)} = 1 - \tfrac{1}{2}(1 - z) - \tfrac{1}{8}(1 - z)^2 + \cdots$$

in the region consisting of the interior of a circle of unit radius about the point $z = 1$. If we try to continue such a function along certain paths in the z-plane (in the case above, any path enclosing the origin), it may happen that upon returning to the original region of definition, we do *not* return to the original values of the function. This leads in a natural way to the construction of Riemann surfaces of the type discussed in Section 6.2.

6.8 RESIDUE THEORY—EVALUATION OF REAL DEFINITE INTEGRALS AND SUMMATION OF SERIES

There is really nothing fundamentally new in this section. All the theorems have been proved; here we just apply them in certain ways to determine the values of some real definite integrals.

The Residue Theorem. The integral of $f(z)$ around a closed contour C containing a finite number n of singular points of $f(z)$ equals the sum of n integrals of $f(z)$ about n circles, each enclosing one (and only one) of the n singular points.

Proof. If we apply Cauchy's theorem to the region shown in Fig. 6.17, we obtain

$$\int_C f(z)\,dz + \int_{C_1} f(z)\,dz + \int_{C_2} f(z)\,dz + \cdots + \int_{C_n} f(z)\,dz = 0.$$

Note that the contours C_j are traversed clockwise in Fig. (6.17). It follows that

$$\oint_C f(z)\,dz = 2\pi i \sum_{j=1}^{n} R_j\,,$$

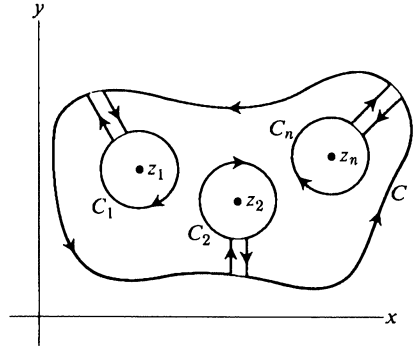

Fig. 6.17

where $R_j = (1/2\pi i) \oint_{C_j} f(z)\, dz$ is called the *residue* at the point z_j. In this equation for R_j the integral sign has its conventional meaning, that is, the contours C_j are traversed counterclockwise.

There is nothing new in this theorem except the name "residue." To compute the residues we shall often use Cauchy's integral formula or the formula for the derivative of an analytic function derived from Cauchy's formula by differentiation.

There is another way to compute residues, however, which is sometimes useful. We expand $f(z)$ in a Laurent series about the singular point z_0:

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} b_n (z - z_0)^{-n} .$$

Now $b_1 = (1/2\pi i) \oint_{C_0} f(z)\, dz = R_0$. It is instructive to check this by integrating both sides of the Laurent series expansion about a circle which includes z_0 but no other singularities. Interchanging summation and integration, which is permissible because these series converge uniformly, we get

$$\oint_{C_0} f(z)\, dz = \sum_{n=0}^{\infty} a_n \oint_{C_0} (z - z_0)^n\, dz + \sum_{n=1}^{\infty} b_n \oint_{C_0} \frac{dz}{(z - z_0)^n} .$$

The first integral vanishes for all n because $(z - z_0)^n$ is analytic. For $n = 1$, the second integral gives $2\pi i b_1$ by Cauchy's integral formula. For $n > 1$, we write $b_n \equiv g(z)$; then by Eq. (6.17)

$$\oint_{C_0} \frac{b_n\, dz}{(z - z_0)^n} = \oint_{C_0} \frac{g(z)\, dz}{(z - z_0)^n} = \frac{2\pi i}{(n - 1)!} g^{(n-1)}(z_0) = 0 ,$$

since $g(z) = b_n = \text{const}$. Thus $(1/2\pi i) \oint_{C_0} f(z)\, dz = b_1 =$ the residue at z_0, so the residue at a point can be found by expanding the function about the point in a Laurent series and picking out the coefficient of the term in $(z - z_0)^{-1}$. There is nothing mysterious about this; it is a simple consequence of Cauchy's integral formula.

Examples.

1. $I = \oint_C e^{1/z}\,dz$, $C =$ unit circle about the origin.

$$e^{1/z} = \sum_{n=0}^{\infty} \frac{1}{n!\,z^n}, \qquad \text{so } b_1 = 1 .$$

Therefore $I = 2\pi i$.

2. $I = \oint_C \dfrac{3z^2 + 2}{z(z+1)}\,dz$,

where C is the circle: $|z| = 3$. The quickest way to do this integral is to write it as

$$\oint_C = \oint_{C_0} + \oint_{C_1} = \oint_{C_0} \frac{(3z^2+2)/(z+1)}{z}\,dz + \oint_{C_1} \frac{(3z^2+2)/z}{(z+1)}\,dz ,$$

where C_0 is a little circle inside C enclosing $z = 0$, but not $z = -1$, and C_1 is a little circle inside C enclosing $z = -1$, but not $z = 0$. Then using Cauchy's integral formula to evaluate both integrals, we obtain $I = 2\pi i(2 - 5) = -6\pi i$.

3. $I = \oint_C \dfrac{3z+2}{z(z+1)^3}\,dz$,

where C is the circle $|z| = 3$. We break up the integral as before:

$$\begin{aligned} I &= \oint_{C_0} \frac{(3z+2)/(z+1)^3}{z}\,dz + \oint_{C_1} \frac{(3z+2)/z}{(z+1)^3}\,dz \\ &= 2\pi i[2] + 2\pi i\left[\frac{f''(z)}{2!}\right]_{z=-1} \end{aligned}$$

where $f(z) = (3z+2)/z$. Therefore $f''(-1) = -4$, so the residue at $z = -1$ is -2, thus $I = 2\pi i(2 - 2) = 0$.

In this last example we evaluated the residue of a function of the form $f(z)/(z - z_0)^n$ at the singular point z_0 by using Cauchy's formula for the derivative of an analytic function. In general, if $f(z)$ is analytic within and on some contour C surrounding z_0, then the residue of the function $f(z)/(z - z_0)^n$ at z_0 is

$$\frac{1}{2\pi i}\oint_C \frac{f(z)}{(z - z_0)^n}\,dz = \frac{f^{n-1}(z_0)}{(n-1)!} .$$

An instructive way to view this formula is to expand $f(z)$ in a Taylor series about z_0. This is possible because $f(z)$ is analytic in some region about z_0, since the singularities are isolated. Then

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \cdots + \frac{f^{n-1}(z_0)(z - z_0)^{n-1}}{(n-1)!} + \cdots .$$

If we now form the quantity $f(z)/(z - z_0)^n$ by dividing both sides by $(z - z_0)^n$, the coefficient of the term in $1/(z - z_0)$, which is just the residue of $f(z)/(z - z_0)^n$, is $f^{(n-1)}(z_0)/(n-1)!$, in agreement with the above.

We now use these techniques to evaluate various definite integrals.

a) $\int_0^{2\pi} R(\cos\theta, \sin\theta)\, d\theta$, where R is a rational function, that is,

$$R = \frac{a_1 \cos\theta + a_2 \sin\theta + a_3 \cos^2\theta + \cdots}{b_1 \cos\theta + b_2 \sin\theta + b_3 \cos^2\theta + b_4 \sin^2\theta + \cdots}.$$

To evaluate this integral, let $z = e^{i\theta}$, so that

$$\cos\theta = \frac{1}{2}\left(z + \frac{1}{z}\right), \qquad \sin\theta = \frac{1}{2i}\left(z - \frac{1}{z}\right),$$

and $d\theta = -i(dz/z)$. The integral becomes

$$-i \oint_C R\left[\frac{1}{2}\left(z + \frac{1}{z}\right), \frac{1}{2i}\left(z - \frac{1}{z}\right)\right]\frac{dz}{z},$$

where C is the unit circle.

Example.

$$I = \int_0^{2\pi} \frac{d\theta}{a + \cos\theta}, \qquad a > 1.$$

$$I = -i \oint_{|z|=1} \frac{1}{\left(a + \frac{z}{2} + \frac{1}{2z}\right)} \frac{dz}{z} = -2i \oint_{|z|=1} \frac{dz}{z^2 + 2az + 1}.$$

The denominator can be factored into $(z - \alpha)(z - \beta)$, where

$$\alpha = -a + (a^2 - 1)^{1/2}, \qquad \beta = -a - (a^2 - 1)^{1/2}.$$

Since $a > 1$, it follows readily that $|\alpha| < 1$ and $|\beta| > 1$. Thus the integrand has one singularity, at $z = \alpha$, within the unit circle, and

$$I = -2i(2\pi i)\frac{1}{\alpha - \beta} = \frac{2\pi}{(a^2 - 1)^{1/2}}.$$

Next we consider integrals of the form:

b) $\int_{-\infty}^{\infty} R(x)\, dx$, where $R(x)$ is a rational function (i.e., a ratio of two polynomials), without poles on the real axis.

If there are no poles on the real axis, then this integral exists if the degree of the denominator of $R(x)$ is at least two units higher than the degree of the numerator. This means that $|R(z)| \to 1/|z^2|$ as $|z| \to \infty$. Now

$$\oint_C R(z)\, dz = \int_{-\rho}^{\rho} R(x)\, dx + \int_{\text{semicircle}} R(z)\, dz$$

(see Fig. 6.18). As $\rho \to \infty$, the closed contour C encloses all the singularities of $R(z)$ in the upper half-plane, so

$$\oint_C R(z)\, dz = 2\pi i \sum_{y>0} \operatorname{Res} R(z),$$

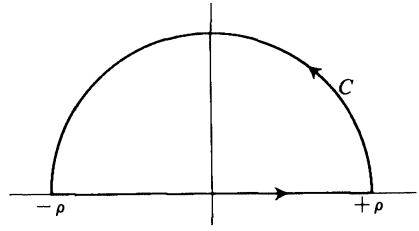

Fig. 6.18

where this notation means to form the sum of all the residues of $R(z)$ in the upper half-plane. Now in the limit of large $|z|$,

$$\left|\int_{\text{semicircle}} R(z)\, dz\right| \leq \int \frac{|\text{const}|\, \rho\, d\theta}{\rho^2} = \frac{|\text{const}|}{\rho},$$

which goes to zero as $\rho \to \infty$. Thus

$$\oint_C R(z)\, dz = \int_{-\infty}^{\infty} R(x)\, dx = 2\pi i \sum_{y>0} \operatorname{Res} R(z)\,.$$

When we say that an integral over a "contour at infinity" vanishes, we really mean, of course, that given any $\epsilon > 0$ there exists a *finite* contour C such that $\left|\oint_C f(z)\, dz\right| < \epsilon$. If this is so, we say that the integral over the contour at ∞ is zero; all contours are really finite.

Example.

$$I = \int_{-\infty}^{\infty} \frac{dx}{(x^2+1)^2}\,.$$

To find the zeros of the denominator in the upper half-plane, we must solve $z^2 = -1$. We get $z = \pm i$. Only the root $+i$ is in the upper half-plane. Therefore

$$I = \oint \frac{dz}{(z+i)^2(z-i)^2} = \frac{2\pi i}{1!}\frac{d}{dz}\left(\frac{1}{(z+i)^2}\right)\Bigg]_{z=i}$$

$$= (2\pi i)\cdot(-2)\cdot\frac{1}{(z+i)^3}\Bigg]_{z=i} = \frac{\pi}{2}\,.$$

c) Another very important class of integrals is $\int_{-\infty}^{\infty} R(x)e^{ix}\, dx$. This is the Fourier integral of the rational function $R(x)$. Its real and imaginary parts determine the integrals:

$$\int_{-\infty}^{\infty} R(x)\cos x\, dx \qquad \text{and} \qquad \int_{-\infty}^{\infty} R(x)\sin x\, dx\,.$$

We retain the assumption that there are no poles on the real axis. Here, too, we consider the integral $\oint_C R(z)e^{iz}\, dz$ over a semicircle. Since $|e^{iz}| = e^{-y} \leq 1$

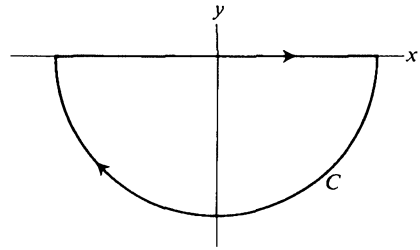

Fig. 6.19

for any point in the upper half-plane, this integral will exist if the rational function $R(z)$ has a zero of at least order two at infinity. Then, just as before, the integral over the infinite semicircle vanishes, leaving

$$\oint_C R(z)e^{iz}\,dz = \int_{-\infty}^{\infty} R(x)e^{ix}\,dx = 2\pi i \sum_{y>0} \operatorname{Res}\left[R(z)e^{iz}\right].$$

Example.

$$I = \int_{-\infty}^{\infty} \frac{e^{ikr}}{k^2+\mu^2}\,dk\,.$$

If $r > 0$, then letting $x = rk$, we find that the integral becomes

$$I = r\int_{-\infty}^{\infty} \frac{e^{ix}}{x^2+(\mu r)^2}\,dx = r\oint \frac{e^{iz}}{z^2+(\mu r)^2}\,dz = 2\pi i r \sum_{y>0} \operatorname{Res}\left(\frac{e^{iz}}{z^2+(\mu r)^2}\right)$$

$$= 2\pi i r \left.\frac{e^{iz}}{z+i\mu r}\right]_{z=i\mu r} = \pi\frac{e^{-\mu r}}{\mu}\,.$$

Note that

$$I = \int_{-\infty}^{\infty} \frac{\cos kr}{k^2+\mu^2}\,dk$$

and hence the result above is actually independent of the sign of r. Therefore

$$\int_{-\infty}^{\infty} \frac{e^{ikr}}{k^2+\mu^2}\,dk = \pi\frac{e^{-\mu|r|}}{\mu}\,.$$

Another way to see this is to compute the integral assuming $r < 0$. Then we must take the contour in the *lower* half-plane, since $|e^{ikr}| = e^{-(\operatorname{Im} k)r}$, which is bounded only for $\operatorname{Im}(k) < 0$, if $r < 0$. As before, for a large enough semicircle (Fig. 6.19),

$$\oint_C = \int_{-\infty}^{\infty} = -2\pi i \sum_{y<0} \operatorname{Res}\,.$$

The minus sign arises because the contour C is traversed clockwise. Thus

$$I = -2\pi i r \left.\frac{e^{iz}}{z-i\mu r}\right]_{z=-i\mu r} = \frac{\pi e^{\mu r}}{\mu}\,,$$

which for $r < 0 = \pi e^{-\mu|r|}/\mu$. Here the residue was computed at the only pole in the lower half-plane, $z = -i\mu r$.

d) There arise integrals of the form $\int_{-\infty}^{\infty} R(x)e^{i\alpha x}\,dx$, where the rational function $R(x)$ has a zero of order one at infinity. The integrals

$$\int_{-\infty}^{\infty} R(x)\cos\alpha x\,dx \qquad \text{and} \qquad \int_{-\infty}^{\infty} R(x)\sin\alpha x\,dx$$

are the real and imaginary parts, respectively, of this integral.

From the preceding discussion, it is not clear that these integrals exist. Jordan's lemma, which we now prove, tells us that they do. Again we assume that there are no poles on the real axis.

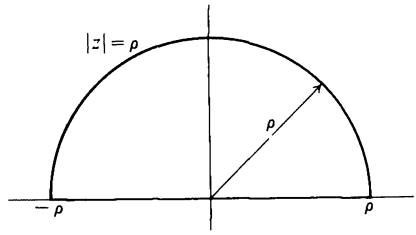

Fig. 6.20

Jordan's Lemma. If as $\rho \to \infty$ (in Fig. 6.20), $|R(z)| \to 0$ uniformly in θ for $0 < \theta < \pi$, then $\lim_{\rho\to\infty} \int_{|z|=\rho} R(z)e^{i\alpha z}\,dz = 0$, for $\alpha > 0$.

Proof. We have almost proved this result in the section on dispersion relations. There, however, the analyticity of $R(z)$ in the upper half-plane was used. Here we use the hypothesis on the uniformity of the approach of $|R(z)|$ to zero. Let $M(\rho)$ be the maximum of $|R(z)|$ on the semicircle $|z| = \rho$. Then the statement that $|R(z)| \to 0$ uniformly means that $|R(z)| \leq M(\rho)$, where $\lim_{\rho\to\infty} M(\rho) = 0$, *independent of* θ.

Let

$$I = \int_{|z|=\rho} R(z)e^{i\alpha z}\,dz\,.$$

Then

$$\begin{aligned} |I| &\leq M(\rho)\int_0^{\pi} |e^{i\alpha(\rho\cos\theta + i\rho\sin\theta)}|\;|\rho i e^{i\theta}|\,d\theta \\ &= \rho M(\rho)\int_0^{\pi} e^{-\alpha\rho\sin\theta}\,d\theta\,. \end{aligned}$$

Treating this integral exactly as in Section 6.6, we obtain

$$|I| < \pi M(\rho)\,\frac{(1 - e^{-\alpha\rho})}{\alpha} \to 0 \qquad \text{as } \rho \to \infty\,,$$

since $\alpha > 0$, and $M(\rho) \to 0$ as $\rho \to \infty$. Thus we arrive at the formula

$$\int_{-\infty}^{\infty} R(x)e^{i\alpha x}\,dx = 2\pi i \sum_{y>0} \operatorname{Res}\,[R(z)e^{i\alpha z}]$$

for $\alpha > 0$. For $\alpha < 0$, the derivation works only in the lower half-plane ($y < 0$), and changes in the above formula must be made accordingly.

Example.

$$
\begin{aligned}
I &= \int \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{k^2+\mu^2}\,d\mathbf{k} \\
&= \int_0^{2\pi} k\,d\phi \int_0^{\pi} k\sin\theta\,d\theta \int_0^{\infty} dk\,\frac{e^{ik|r|\cos\theta}}{k^2+\mu^2} \\
&= 2\pi\int_{-1}^{1} dt \int_0^{\infty} dk\,\frac{k^2 e^{ik|r|t}}{k^2+\mu^2} \\
&= \frac{2\pi}{i|r|}\int_0^{\infty} dk\,\frac{k(e^{ik|r|}-e^{-ik|r|})}{k^2+\mu^2} \\
&= \frac{2\pi}{i|r|}\int_{-\infty}^{\infty} dk\,\frac{ke^{ik|r|}}{k^2+\mu^2}\,.
\end{aligned}
$$

Jordan's lemma tells us that this integral for complex k vanishes over the infinite semicircle. Therefore

$$
I = \frac{2\pi}{i|r|}\,2\pi i\sum_{y>0}\operatorname{Res}\frac{ke^{ik|r|}}{k^2+\mu^2} = \frac{4\pi^2}{|r|}\,\frac{i\mu e^{i\cdot i\mu|r|}}{2i\mu} = \frac{2\pi^2}{|r|}\,e^{-\mu|r|}\,, \tag{6.51}
$$

since there is only one pole in the upper half-plane ($k = +i\mu$).

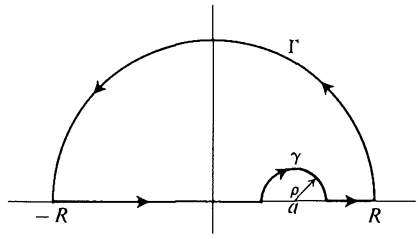

Fig. 6.21

e) We have not as yet allowed the integrand to have poles on the real axis. If $Q(z)$ is a meromorphic function in the upper half-plane, if it only has poles of order one (i.e., simple poles) on the real axis, and if $Q(z)$ behaves at infinity like any of the integrands of class b, c, or d, then the techniques of these cases can be extended to evaluate integrals of the form

$$
P\int_{-\infty}^{\infty} Q(x)\,dx,
$$

where the P stands for Cauchy's principal value, which arises because of the presence of the poles on the real axis.

Let $Q(z)$ have a single simple pole on the real axis at $z = a$. Consider the indented contour C (Fig. 6.21) consisting of the small semicircle γ about point a, the large semicircle Γ about the origin, and the two straight line segments on the real axis from $-R$ to $a - \rho$ and from $a + \rho$ to R. We take γ small enough so it encloses only the pole at a; Γ is taken large enough to enclose all

the poles in the upper half-plane and large enough so the integral over Γ approaches 0 as $R \to \infty$. We have

$$\left(\int_\Gamma + \int_{-R}^{a-\rho} + \int_\gamma + \int_{a+\rho}^{R}\right) Q(z)\, dz = 2\pi i \sum_{y>0} \operatorname{Res} Q(z) .$$

Taking the limit as $R \to \infty$, we get

$$\left[\lim_{R\to\infty}\left(\int_{-R}^{a-\rho} + \int_{a+\rho}^{R}\right) + \int_\gamma\right] Q(z)\, dz = P\int_{-\infty}^{\infty} Q(x)\, dx + \int_\gamma Q(z)\, dz$$
$$= 2\pi i \sum_{y>0} \operatorname{Res} Q(z) .$$

We now consider $\int_\gamma Q(z)\, dz$. On γ, $z = a + \rho e^{i\theta}$, so that

$$\int_\gamma Q(z)\, dz = \int_\pi^0 Q(a + \rho e^{i\theta}) \rho e^{i\theta} i\, d\theta .$$

Since $Q(z)$ has a simple pole at $z = a$, it contains the factor $(z - a)^{-1}$. We may therefore write $Q(z) = \phi(z)/(z - a) + \psi(z)$, where $\phi(z)$ and $\psi(z)$ are analytic at and near $z = a$. Clearly $\psi(z)$ does not contribute to the integral over γ as $\rho \to 0$, so since $z - a = \rho e^{i\theta}$, we have

$$\int_\gamma Q(z)\, dz = \int_\pi^0 \phi(a + \rho e^{i\theta}) i\, d\theta .$$

We now expand ϕ in a Taylor series about a (it is analytic there):

$$\phi(a + \rho e^{i\theta}) = \phi(a) + \text{terms in } \rho .$$

Therefore, letting $\rho \to 0$, we get

$$\int_\gamma Q(z)\, dz = \int_\pi^0 \phi(a) i\, d\theta = -i\pi\phi(a) .$$

Now $\phi(a)$ is the residue of $Q(z) = \phi(z)/(z - a)$ at $z = a$, so the final answer may be written as

$$P\int_{-\infty}^{\infty} Q(x)\, dx = 2\pi i \sum_{y>0} \operatorname{Res} Q(z) + \pi i \sum_{y=0} \operatorname{Res} Q(z) ,$$

where $\sum_{y=0} \operatorname{Res} Q(z)$ denotes the sum of the residues of $Q(z)$ at each of its simple poles on the real axis (generalizing from one to several simple poles).

Example.

$$I = \int_{-\infty}^{\infty} \frac{\sin x}{x}\, dx .$$

The value of this integral may be determined from the following simple result:

$$P\int_{-\infty}^{\infty} \frac{e^{ix}}{x}\, dx = \pi i \sum_{y=0} \operatorname{Res}\left(\frac{e^{iz}}{z}\right) = \pi i .$$

Here the only pole is on the real axis at $z = x = 0$.

Equating real and imaginary parts, we have

$$P\int_{-\infty}^{\infty} \frac{\cos x}{x}\, dx = 0\,,$$

which is trivial because the integrand is an odd function, and

$$P\int_{-\infty}^{\infty} \frac{\sin x}{x}\, dx = \pi\,,$$

a result derived previously from Hilbert transform theory. In this formula the P is superfluous because the integrand has no pole. Many integrals involving $\sin x$ or $\cos x$ may be evaluated, as we have done here, by replacing the trigonometric function with e^{iz}, and later taking real and imaginary parts.

f) Next we consider integrals of the form $\int_0^\infty x^{\lambda-1} R(x)\, dx$, where $R(z)$ is rational, analytic at $z = 0$, and has no poles on the positive real axis, and where $|z^\lambda R(z)| \to 0$ uniformly as $|z| \to 0$ and as $|z| \to \infty$. Since the case of integral λ can be handled by the methods described earlier, we will assume that λ is not equal to an integer.

This problem involves branch points and branch cuts because $z^{\lambda-1}$ is not in general, a single-valued function. The power function $z^{\lambda-1}$ has a branch point at the origin. Let $z^{\lambda-1}$ denote the following branch of the power function:

$$z^{\lambda-1} = \exp[\,(\lambda - 1)\log z] = \exp[\,(\lambda - 1)(\log r + i\theta)]\,,$$

where $0 < \theta < 2\pi$, $r > 0$. Then

$$z^{\lambda-1} = r^{\lambda-1}e^{i(\lambda-1)\theta}\,, \qquad 0 < \theta < 2\pi,\, r > 0\,.$$

The branch cut has been chosen to be the positive real axis. For $\theta = 0$, the power function has the value

$$z^{\lambda-1} = r^{\lambda-1} = x^{\lambda-1}\,.$$

Now consider the contour integral $\oint_{C'} z^{\lambda-1}R(z)\, dz$, where the closed contour C' is shown in Fig. 6.22. Here C' consists of a small circle about $z = 0$, whose radius will later be shrunk to zero, a large circle whose radius will later be expanded to ∞, and two integrals along the positive real axis in opposite directions and on opposite sides of the branch cut. Since the integrand is discontinuous along the branch cut, these two integrals will not cancel. To do the integral, the phase of $z^{\lambda-1}$ must be prescribed everywhere. We have done this above by defining it to be 0 on the positive real axis so that $z^{\lambda-1} = x^{\lambda-1}$ there, which is the usual convention. Therefore, on the line just below the real axis, $z^{\lambda-1} = x^{\lambda-1}e^{i2\pi(\lambda-1)}$. The integrals over the little and big circles vanish, because $|z^\lambda R(z)| \to 0$ as $|z| \to 0$, and as $|z| \to \infty$. Therefore

$$\oint_{C'} z^{\lambda-1}R(z)\, dz = -\int_0^\infty e^{2\pi i(\lambda-1)}x^{\lambda-1}R(x)\, dx + \int_0^\infty x^{\lambda-1}R(x)\, dx\,,$$

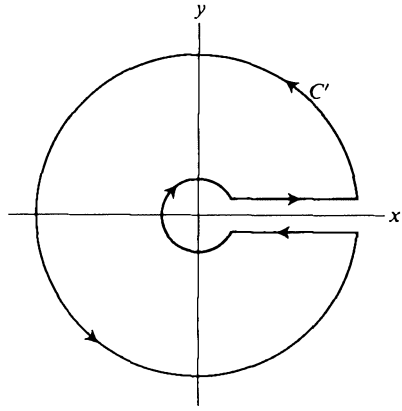

Fig. 6.22

where the first integral on the right is the contribution from the line below the positive x-axis and the second is the contribution from the line above the x-axis. Together these integrals give

$$[1 - e^{2\pi i(\lambda-1)}]\int_0^\infty x^{\lambda-1}R(x)\,dx = \left(\frac{-2i\sin\pi\lambda}{e^{-\pi i\lambda}}\right)\int_0^\infty x^{\lambda-1}R(x)\,dx\,.$$

Since

$$\oint_{C'} z^{\lambda-1}R(z)\,dz = 2\pi i \sum_{\text{inside } C'} \operatorname{Res}\,[z^{\lambda-1}R(z)]\,,$$

we obtain

$$\begin{aligned}\int_0^\infty x^{\lambda-1}R(x)\,dx &= \frac{-\pi e^{-\pi i\lambda}}{\sin\pi\lambda}\sum_{\text{inside } C'} \operatorname{Res}\,[z^{\lambda-1}R(z)] \qquad (6.52)\\ &= \frac{\pi(-1)^{\lambda-1}}{\sin\pi\lambda}\sum_{\text{inside } C'} \operatorname{Res}\,[z^{\lambda-1}R(z)]\,,\end{aligned}$$

since $e^{-i\pi\lambda} = (-1)^\lambda$. This is as far as we can go without choosing a specific function $R(z)$. A very simple example of this type is the following integral:

$$I = \int_0^\infty \frac{x^{\lambda-1}}{1+x}\,dx\,, \qquad 0 < \lambda < 1\,.$$

After evaluating it, we shall use it to prove some results involving the beta and gamma functions.

First since $0 < \lambda < 1$, $|z^\lambda/(1+z)| \to 0$ as $|z| \to \infty$ and as $z \to 0$. Therefore

$$\int_0^\infty \frac{x^{\lambda-1}}{1+x}\,dx = \frac{\pi(-1)^{\lambda-1}}{\sin\pi\lambda}\operatorname{Res}\left(\frac{z^{\lambda-1}}{1+z}\right)_{z=-1} = \frac{\pi}{\sin\pi\lambda}.$$

The beta function $\beta(x, y)$ is defined as

$$\beta(x, y) \equiv \int_0^1 t^{x-1}(1-t)^{y-1}\,dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}\,, \qquad x, y > 0\,, \qquad (6.53)$$

where $\Gamma(x)$ is the gamma function:

$$\Gamma(x) = \int_0^\infty t^{x-1}e^{-t}\,dt\,. \tag{6.54}$$

The proof of the identity in Eq. (6.53) may be found in many books on analysis. We note that

$$\beta(x, 1-x) = \Gamma(x)\Gamma(1-x) = \int_0^1 t^{x-1}(1-t)^{-x}\,dt\,, \qquad \text{if } 0 < x < 1.$$

If in the last integral we make the change of variable

$$t = \frac{u}{1+u} = 1 - \frac{1}{1+u}\,,$$

it becomes

$$\int_0^\infty \frac{u^{x-1}}{1+u}\,du\,,$$

which is precisely the integral we have just evaluated. Thus

$$\beta(x, 1-x) = \Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin \pi x}\,, \qquad \text{for } 0 < x < 1\,. \tag{6.55}$$

This formula may be extended by analytic continuation to all z in the complex plane. In particular, for $x = \frac{1}{2}$, we have $[\Gamma(\frac{1}{2})]^2 = \pi$. Therefore

$$\pi^{1/2} = \Gamma(\tfrac{1}{2}) \equiv \int_0^\infty t^{-1/2}e^{-t}\,dt = 2\int_0^\infty e^{-u^2}\,du = \int_{-\infty}^\infty e^{-u^2}\,du\,, \tag{6.56}$$

where we have made the substitution $t = u^2$.

We close this section on residue theory with an illustration of how series can be summed by contour integration. The result we shall obtain depends on the fact that $\pi \cot \pi z$ has poles of order one at the zeros of $\sin \pi z$, (that is, at $z = n$, $n = 0, \pm 1, \pm 2, \cdots$), and the fact that the residue at each of these poles is 1.

Theorem. Let $f(z)$ be a meromorphic function and let C be a contour which encloses the zeros of $\sin \pi z$, located at $z = \rho, \rho + 1, \cdots, n$. If we assume that the poles of $f(z)$ and $\sin \pi z$ are distinct, then

$$\sum_{m=\rho}^{n} f(m) = \frac{1}{2\pi i}\oint_C \pi \cot \pi z\, f(z)\,dz - \sum_{\substack{\text{poles of } f(z)\\ \text{inside } C}} \operatorname{Res}\,[\pi \cot (\pi z)\, f(z)\,]\,. \tag{6.57}$$

Proof.

$$\oint_C \pi \cot \pi z f(z)\,dz = 2\pi i \sum \text{(Residues at all the poles of the integrand)}$$

$$= 2\pi i\left[\sum_{m=\rho}^{n} f(m) + \sum_{\substack{\text{poles of } f(z)\\ \text{inside } C}} \operatorname{Res}\,[\pi \cot \pi z\, f(z)\,]\right]. \quad \text{QED}$$

Example. We shall use this theorem to establish the equivalence of Langevin's function $(\coth x - 1/x)$ and the sum

$$\sum_{n=1}^{\infty} \frac{2x}{x^2 + n^2\pi^2}.$$

We shall then use this result to derive an infinite *product* expression for $\sin\theta$, which is of considerable interest in its own right.

Letting $f(z) = 2x/(x^2 + z^2\pi^2)$, and using the formula above, we obtain

$$\sum_{m=-N}^{N} \frac{2x}{x^2 + m^2\pi^2} = \frac{1}{2\pi i}\oint_C \pi \cot \pi z\, f(z)\, dz - \sum_{\substack{\text{poles of } f(z)\\ \text{inside } C}} \operatorname{Res}\left[\pi \cot \pi z\, f(z)\right],$$

where C is a closed contour, say, a rectangle, enclosing the points $z = -N, -N+1, \cdots, 0, 1, \cdots, N-1, N$. Now let the length and width of the rectangle C approach ∞. As this happens,

$$\left|\frac{1}{2\pi i}\oint_C \pi \cot \pi z\, f(z)\, dz\right| \leq \tfrac{1}{2}\oint_C |\cot \pi z| \left|\frac{2x}{x^2 + z^2\pi^2}\right| |dz| \to 0.$$

To see that this integral vanishes as $z \to \infty$, we observe that

$$|\cot \pi z| = \frac{|\cos \pi z|}{|\sin \pi z|} = \left(\frac{\cos^2 \pi x + \sinh^2 \pi y}{\sin^2 \pi x + \sinh^2 \pi y}\right)^{1/2}.$$

Now, arbitrarily high accuracy can be achieved in summing the series by choosing the rectangle so that its vertical sides cross the x-axis at a large enough half-integer, for example $(10^{97} + \frac{1}{2})$, where $\cos \pi(10^{97} + \frac{1}{2}) = 0$ and $\sin \pi(10^{97} + \frac{1}{2}) = 1$. Then, over these sides of the rectangle

$$|\cot \pi z| = \left|\left(\frac{\sinh^2 \pi y}{1 + \sinh^2 \pi y}\right)^{1/2}\right| = |\tanh \pi y| \leq 1.$$

Over the horizontal sides of the rectangle, $\lim_{z\to\infty} |\cot \pi z| = 1$. Thus the integrand goes as $|1/z^2|$ as $|z| \to \infty$, and the integral vanishes.

If we take an infinite rectangular contour, then

$$\sum_{m=-\infty}^{\infty} \frac{2x}{x^2 + m^2\pi^2} = -\operatorname{Res}\left[\frac{(\pi \cot \pi z)\, 2x}{x^2 + z^2\pi^2}\right]_{z=\pm ix/\pi}$$

$$= -\frac{2x}{\pi}\left[\frac{\cot \pi(ix/\pi)}{2ix/\pi} + \frac{\cot \pi(-ix/\pi)}{-2ix/\pi}\right] = 2i \cot ix = 2 \coth x.$$

Therefore

$$2\sum_{m=1}^{\infty} \frac{2x}{x^2 + m^2\pi^2} + \frac{2}{x} = 2 \coth x$$

or

$$\coth x - \frac{1}{x} = \sum_{m=1}^{\infty} \frac{2x}{x^2 + m^2\pi^2}, \tag{6.58}$$

which establishes the result stated at the outset.

Aside from illustrating this summation technique, this particular result is not of much interest. However, if we integrate both sides from 0 to x, we get

$$\sum_{m=1}^{\infty} \ln\left(1 + \frac{x^2}{m^2\pi^2}\right) = \ln\left(\frac{\sinh x}{x}\right).$$

Hence

$$\ln\left(\frac{\sinh x}{x}\right) = \sum_{m=1}^{\infty} \ln\left(1 + \frac{x^2}{m^2\pi^2}\right) = \ln \prod_{m=1}^{\infty}\left(1 + \frac{x^2}{m^2\pi^2}\right),$$

so

$$\frac{\sinh x}{x} = \prod_{m=1}^{\infty}\left(1 + \frac{x^2}{m^2\pi^2}\right). \tag{6.59}$$

We may extend this result to all z in the complex plane by analytic continuation. Then setting $x = i\theta$ (θ real), we obtain

$$\sin\theta = \theta \prod_{n=1}^{\infty}\left(1 - \frac{\theta^2}{n^2\pi^2}\right). \tag{6.60}$$

This infinite product formula displays *explicitly* all the zeros of $\sin\theta$. It represents the complete factorization of the Taylor series. It can, in fact, be taken as the definition of the sine function.

By equating coefficients of the θ^3 term of both sides of the above equation, we obtain a useful sum:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}. \tag{6.61}$$

This is a special value of the Riemann zeta function,

$$\zeta(z) = \sum_{n=1}^{\infty} \frac{1}{n^z}. \tag{6.62}$$

There are many other special tricks for evaluating integrals and summing series. We have surveyed only the principal techniques, but the reader will have no trouble understanding any particular evaluation procedure he may encounter if he understands this section. Problem 31 provides applications of all the basic techniques of residue calculus developed above.

6.9 APPLICATIONS TO SPECIAL FUNCTIONS AND INTEGRAL REPRESENTATIONS

In this section we use the formula for the derivative of an analytic function to find generating functions for certain special functions from their Rodrigues formulas. Also, we shall derive integral representations for Bessel functions and Legendre polynomials.

Bessel Functions

The function $e^{(1/2)z(w-1/w)}$ is analytic everywhere in the w-plane except at $w = 0$, so it can be expanded in a Laurent series in any annulus $R_2 < |w| < R_1$, no

matter how small $R_2(>0)$ or how large R_1. Denoting the expansion coefficients by $J_n(z)$, we have

$$e^{(1/2)z(w-1/w)} = \sum_{n=-\infty}^{\infty} J_n(z) w^n . \tag{6.63}$$

The expansion coefficients are functions of the complex variable z. The notation chosen for them anticipates the fact that they will turn out to be Bessel functions of integral order. In other words, $e^{(1/2)z(w-1/w)}$ is a generating function for Bessel functions.

We now prove that the functions $J_n(z)$ do satisfy Bessel's differential equation:

$$z^2 \frac{d^2 J_n}{dz^2} + z \frac{dJ_n}{dz} + (z^2 - n^2) J_n = 0 . \tag{6.64}$$

Along the way to this result we shall find an integral representation for Bessel functions and also an explicit formula for them.

By Laurent's theorem,

$$J_n(z) = \frac{1}{2\pi i} \oint_C \frac{f(w)}{(w - w_0)^{n+1}} dw , \tag{6.65}$$

where C is any closed contour in the annulus. We take C to be the unit circle: $w = e^{i\theta}$. Then in Eq. (6.65), $w_0 = 0$ and $f(w) = e^{(1/2)z(w-1/w)}$. On the unit circle $w - 1/w = 2i \sin\theta$. Therefore

$$\begin{aligned} J_n(z) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{iz\sin\theta} e^{-in\theta} d\theta = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-i(n\theta - z\sin\theta)} d\theta \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(n\theta - z\sin\theta) d\theta - \frac{i}{2\pi} \int_{-\pi}^{\pi} \sin(n\theta - z\sin\theta) d\theta . \end{aligned}$$

But the second integral is zero because the integrand is odd in θ. Since the first integrand is even in θ,

$$J_n(z) = \frac{1}{\pi} \int_0^{\pi} \cos(n\theta - z\sin\theta) d\theta . \tag{6.66}$$

This is an integral representation for the function $J_n(z)$, which we now prove is a solution of Bessel's equation by substituting it into Eq. (6.64). The verification is trickier than one might expect. We therefore resist the temptation to "leave it to the reader." He is, of course, welcome to have a try at it before reading further. We have

$$J_n'(z) = \frac{1}{\pi} \int_0^{\pi} \sin\theta \sin(n\theta - z\sin\theta) d\theta \tag{6.67a}$$

$$= -\frac{1}{\pi} \cos\theta \sin(n\theta - z\sin\theta) \Big]_0^{\pi}$$

$$+ \frac{1}{\pi} \int_0^{\pi} \cos\theta \cos(n\theta - z\sin\theta)(n - z\cos\theta) d\theta , \tag{6.67b}$$

where we have integrated by parts. The first term is zero. Differentiating Eq. (6.67a) again, we have

$$J_n''(z) = -\frac{1}{\pi}\int_0^\pi \sin^2\theta \cos(n\theta - z\sin\theta)d\theta .$$

We now form the quantity $z^2(J_n'' + J_n) + zJ_n' - n^2J_n$, using Eq. (6.67b) for $J_n'(x)$. It must be identically zero if the $J_n(z)$ are Bessel functions. We get

$$\frac{1}{\pi}\int_0^\pi \cos(n\theta - z\sin\theta)[z^2 - z^2\sin^2\theta + z\cos\theta(n - z\cos\theta) - n^2]\,d\theta$$
$$= \frac{-n}{\pi}\int_0^\pi [\cos(n\theta - z\sin\theta)(n - z\cos\theta)]\,d\theta .$$

The function $\sin(n\theta - z\sin\theta)$ is an antiderivative of the integrand. It vanishes at both 0 and π, so the proof is complete: the $J_n(z)$ are solutions of Bessel's equation. It is easy to show that

$$\begin{aligned} J_0(0) &= 1 , \qquad J_n(0) = 0 \quad \text{for } n \neq 0 , \\ J_1'(0) &= \tfrac{1}{2} , \qquad J_n'(0) = 0 \quad \text{for } n \neq 1 , \end{aligned} \tag{6.68}$$

so these functions are indeed that solution of Bessel's equation, known as Bessel functions of nth (integral) order of the *first* kind (which are analytic at the origin).

Usually, the solution of Bessel's equation is expressed as an infinite series, not as an integral. We now derive the infinite series solution from the generating function:

$$\begin{aligned} e^{(1/2)z(w-1/w)} &= e^{(1/2)zw}e^{-z/2w} \\ &= \sum_{r=0}^{\infty}\frac{(z/2)^r w^r}{r!}\sum_{m=0}^{\infty}\frac{(-z/2)^m w^{-m}}{m!} = \sum_{n=-\infty}^{\infty} w^n J_n(z) . \end{aligned}$$

To obtain $J_n(z)$, the coefficient of the term in $w^n (n \geq 0)$, we multiply each term in w^{-m} in the second series by the term w^{n+m} in the first series, and then sum over all m:

$$\begin{aligned} J_n(z)w^n &= \sum_{m=0}^{\infty}\frac{(z/2)^{n+m}w^{n+m}(-z/2)^m w^{-m}}{(n+m)!\,m!} \\ &= \sum_{m=0}^{\infty}\left[\frac{(-1)^m(z/2)^{2m+n}}{(n+m)!\,m!}\right]w^n . \end{aligned}$$

Thus

$$J_n(z) = \sum_{m=0}^{\infty}\frac{(-1)^m(z/2)^{2m+n}}{(n+m)!\,m!} . \tag{6.69}$$

This is the infinite series solution for integral $n \geq 0$. The reader may show that, for integral n,

$$J_{-n}(z) = (-1)^n J_n(z) . \tag{6.70}$$

Thus these solutions are linearly dependent, and there must exist another set of linearly independent solutions. They are called the Neumann functions, and are denoted by $N_n(z)$. A whole family of equations and solution functions may be constructed from Bessel's equations and Bessel functions by permitting the index n to be nonintegral and half-integral. Also, various linear combinations of Bessel and Neumann functions of real and imaginary argument give rise to the modified Bessel functions and the Hankel functions. The spherical Bessel functions (which we discuss in Chapter 7) are still another set of functions that are defined in terms of the Bessel functions. Each of these sets of functions is a Sturm-Liouville system. It is not our purpose to provide an exhaustive (and exhausting) account of them, but merely to derive and discuss the fundamental results so the reader can dig the particular facts he needs for his work out of a treatise on the subject.

Legendre polynomials

Previously we have shown that the Legendre polynomials, defined by the Rodrigues formula

$$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2 - 1)^n ,$$

are an orthogonal set on the interval $[-1, 1]$ and satisfy Legendre's equation:

$$(1 - x^2)P_n'' - 2xP_n' + n(n + 1)P_n = 0 .$$

It was mentioned in Section 1.8, but not proved, (except for P_0, P_1, and P_2), that a generating function for the Legendre polynomials is

$$\frac{1}{(1 - 2xt + t^2)^{1/2}} = \sum_{n=0}^{\infty} P_n(x)t^n , \qquad 0 < t < 1 , \qquad |x| \leq 1 . \quad (6.71)$$

In the comparison of this equation with Eq. (1.101), the reader will note that we have set $t \equiv r'/r$ and $x \equiv \cos\theta$. We now prove this relation for all n. We begin by expressing the Rodrigues formula as a contour integral by making use of the formula for the nth derivative of an analytic function:

$$P_n(x) = \frac{(-1)^n}{2^n n!}\frac{d^n}{dx^n}(1 - x^2)^n = \frac{(-1)^n}{2^n}\frac{1}{2\pi i}\oint_C \frac{(1 - z^2)^n}{(z - x)^{n+1}}\,dz . \quad (6.72)$$

Here we have used the fact that the function $(1 - z^2)^n$ is entire (C is a closed contour that encloses the point x). This integral representation of the Legendre polynomials is known as Schläfli's integral formula. If we now form the series $S = \sum_{n=0}^{\infty} P_n(x)t^n$, using this integral representation for $P_n(x)$, and interchange the order of summation and integration, we get

$$S = \frac{1}{2\pi i}\oint_C \frac{dz}{z - x}\sum_{n=0}^{\infty}\frac{(-1)^n}{2^n}t^n\left(\frac{1 - z^2}{z - x}\right)^n .$$

This is a simple geometric series, which converges and is easily summed if

$$\left|\frac{t}{2}\frac{1 - z^2}{z - x}\right| < 1 .$$

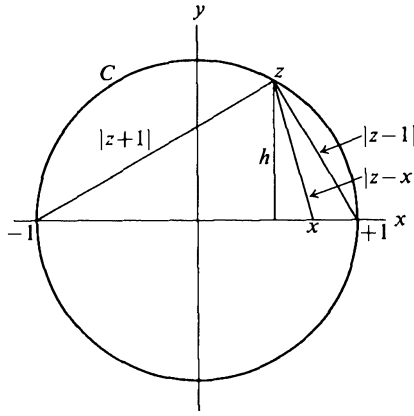

Fig. 6.23

We must show, however, that a contour C, enclosing the point x, can always be chosen so that this is true for all x and t in their prescribed ranges. Let C be the unit circle; z is a point anywhere on C, and x is on the real axis inside or on C as shown in Fig. 6.23. The area A of the right triangle with vertices at -1, z, and $+1$ is given by

$$A = \tfrac{1}{2}\,|z+1|\,|z-1| = \tfrac{1}{2}\,|1-z^2|\,.$$

It is also given by $\frac{1}{2}(2h) = h$. But $|z-x| \geq h$ for all z and x, so

$$|z-x| \geq h = A = \tfrac{1}{2}\,|1-z^2| \Longrightarrow \left|\tfrac{1}{2}\cdot\frac{1-z^2}{z-x}\right| \leq 1\,.$$

Consequently,

$$\left|\frac{t}{2}\,\frac{1-z^2}{z-x}\right| < 1$$

since $t < 1$. A purely algebraic proof is also possible, but this geometric one is simpler. We now sum the geometric series to obtain

$$S = \frac{1}{2\pi i}\oint_C \frac{dz}{z-x}\,\frac{1}{1-\left[\dfrac{-t}{2}\,\dfrac{(1-z^2)}{(z-x)}\right]} = -\frac{1}{2\pi i}\oint_C \frac{2/t\,dz}{z^2-\dfrac{2}{t}z-\left(1-\dfrac{2}{t}x\right)}\,.$$

The denominator has two roots;

$$z_\pm = \frac{1}{t} \pm \left(\frac{1}{t^2}-\frac{2x}{t}+1\right)^{1/2} = \frac{1}{t}\left[1 \pm (1-2xt+t^2)^{1/2}\right].$$

Remembering that $0 < t < 1$ and $-1 \leq x \leq 1$, it is easy for us to show that $z_+ > 1$ and $-1 < z_- < 1$, so z_- is the root enclosed by the contour C. Evaluating the residue at z_-, we have

$$S = \sum_{n=0}^{\infty} P_n(x)t^n = \frac{-1}{2\pi i}\,\frac{2}{t}\,2\pi i\,\frac{1}{z_- - z_+} = \frac{1}{(1-2xt+t^2)^{1/2}}\,. \qquad \text{QED}$$

Table 6.1

PROPERTIES OF ORTHOGONAL FUNCTIONS ARISING FROM STURM-LIOUVILLE SYSTEMS

Name and physical application	Rodrigues formula	Generating function
Legendre polynomials: 1) Multipole expansion 2) ∇^2 in spherical coordinates	$P_n(x) = \frac{1}{2^n n!}\frac{d^n}{dx^n}(x^2-1)^n$	$\sum_{n=0}^{\infty} P_n(x)t^n = \frac{1}{\sqrt{1-2xt+t^2}}$ $(0 < t < 1)$
Hermite polynomials: Quantum oscillator	$H_n(x) = (-1)^n e^{x^2}\frac{d^n e^{-x^2}}{dx^n}$	$\sum_{n=0}^{\infty}\frac{H_n(x)}{n!}t^n = e^{-t^2+2tx}$ $(t > 0)$
Laguerre polynomials: Hydrogen atom (radial equation)	$L_n(x) = e^x\frac{d^n}{dx^n}x^n e^{-x}$	$\sum_{n=0}^{\infty}\frac{L_n(x)}{n!}t^n = \frac{e^{-xt/(1-t)}}{1-t}$ $(0 < t < 1)$
	Series representation	
Bessel's function (of integral order) ∇^2 in cylindrical coordinates	$J_n(x) = \sum_{m=0}^{\infty}\frac{(-1)^m(x/2)^{2m+n}}{(n+m)!\,m!}$	$\sum_{n=-\infty}^{\infty} J_n(x)t^n = e^{(1/2)x(t-1/t)}$ $(t > 0)$
Trigonometric functions: Classical oscillator	$f_n = \sin nx = \sum_{m=0}^{\infty}(-1)^m\frac{(nx)^{2m+1}}{(2m+1)!}$ $g_n = \cos nx = \sum_{m=0}^{\infty}(-1^m)\frac{(nx)^{2m}}{(2m)!}$	

Differential equation	Differential equation in Sturm-Liouville form	Orthonormality
$(1-x^2)P_n'' - 2xP_n' + n(n+1)P_n = 0$	$\frac{d}{dx}((1-x^2)P_n') + n(n+1)P_n = 0$	$\int_{-1}^{1} P_n P_m \, dx = \delta_{nm} \frac{2}{2n+1}$
$H_n'' - 2xH' + 2nH_n = 0$	$\frac{d}{dx}(e^{-x^2}H_n') + 2ne^{-x^2}H_n = 0$	$\int_{-\infty}^{\infty} H_n H_m e^{-x^2} \, dx = \delta_{nm}\sqrt{\pi}2^n n!$
$xL_n'' + (1-x)L_n' + nL_n = 0$	$\frac{d}{dx}(xe^{-x}L_n') + ne^{-x}L_n = 0$	$\int_{0}^{\infty} L_n L_m e^{-x} \, dx = \delta_{nm}(n!)^2$
$x^2J_n'' + xJ_n' + (x^2-n^2)J_n = 0$	$\frac{d}{dx}(xJ_n') + \left(x - \frac{n^2}{x}\right)J_n = 0$	See Problem 5.23
$f_n'' + n^2 f_n = 0$ $g_n'' + n^2 g_n = 0$	$\frac{d}{dx}(f_n') + n^2 f_n = 0$ $\frac{d}{dx}(g_n') + n^2 g_n = 0$	$\int_{-\pi}^{\pi} f_n f_m \, dx = \delta_{nm}\pi$ $\int_{-\pi}^{\pi} g_n g_m \, dx = \delta_{nm}\pi$ $\int_{-\pi}^{\pi} f_n g_m \, dx = 0$

In exactly the same way, the generating functions for the Hermite polynomials and the Laguerre polynomials can be derived from their Rodrigues formulas (see Problems 36 and 37).

Let us return to Schläfli's integral representation of the Legendre polynomials:

$$P_n(x) = \frac{1}{2\pi i}\oint_C \frac{(z^2-1)^n}{2^n(z-x)^{n+1}}\,dz\,. \tag{6.73}$$

It can be proved directly from this integral representation that the $P_n(x)$ satisfy Legendre's differential equation. But we have already proved that the $P_n(x)$ as given by Rodrigues' formula satisfy Legendre's equation, so we omit this proof. Here we derive another integral representation: a real integral.

For the contour C, we take a circle about x of radius $|(x^2-1)^{1/2}|$. Then for any point z on C, we may put $z = x + (x^2-1)^{1/2}e^{i\theta}$, where θ increases from $-\pi$ to π. Making this substitution in Schläfli's formula, we obtain

$$P_n(x) = \frac{1}{2\pi i}\int_{-\pi}^{\pi}\left[\frac{(x-1+(x^2-1)^{1/2}e^{i\theta})(x+1+(x^2-1)^{1/2}e^{i\theta})}{2(x^2-1)^{1/2}e^{i\theta}}\right]^n i\,d\theta\,,$$

where we have written $(z^2-1)^n$ in the form $[\,(z-1)(z+1)\,]^n$ before substituting $z = x + (x^2-1)^{1/2}e^{i\theta}$. Now everything in square brackets above simplifies after a little algebra to $x + (x^2-1)^{1/2}\cos\theta$, an even function of θ. Thus we have a real integral representation of the Legendre polynomials due to Laplace:

$$P_n(x) = \frac{1}{\pi}\int_0^{\pi}[x + (x^2-1)^{1/2}\cos\theta]^n\,d\theta\,, \qquad \text{for } |x| \le 1\,. \tag{6.74}$$

Note how obvious it is from this representation that $P_n(1) = 1$, and $P_n(-1) = (-1)^n$. It is instructive to compute P_0, P_1, and P_2 from this formula. Although $(x^2-1)^{1/2}$ is pure imaginary for $|x| < 1$, all the terms in $(x^2-1)^{1/2}$ vanish when the integral is performed because $\int_0^{\pi}\cos^m\theta\,d\theta = 0$ for odd m.

We conclude this section with Table 6.1. It summarizes some of the more useful information concerning the special functions of mathematical physics.

PROBLEMS

1. Let

$$f(z) = \frac{xy^2(x+iy)}{x^2+y^2} \qquad \text{for } z \ne 0\,, \quad f(0) = 0\,.$$

Determine where, if anywhere, this function is a) differentiable, b) analytic.

2. Show that the complex numbers z_1, z_2, z_3 lie on a straight line if and only if $(z_1 - z_3)/(z_2 - z_3)$ is a real number.
[*Hint*: This problem can be done very simply if one thinks of the complex numbers geometrically (as vectors). It can be done algebraically also, but the geometric proof is certainly easier.]

3. Prove that $u = \sin x \cosh y + 2 \cos x \sinh y + x^2 - y^2 + 4xy$ is a harmonic function and find a conjugate harmonic function v. Find a complex function of the complex variable z such that $f(z) = u + iv$.
4. a) Determine all the values of the constants a, b, c, d for which the polynomial $u(x, y) = ax^3 + bx^2y + cxy^2 + dy^3$ is harmonic, i.e., satisfies the two-dimensional Laplace equation in the entire plane.
   b) Find a harmonic conjugate, $v(x, y)$, of $u(x, y)$.
   c) Find an analytic function $f(z) = u(x, y) + iv(x, y)$, where $z = x + iy$.
5. A complex function $E^*(z)$ is defined by $E^*(z) = E_x(x, y) - iE_y(x, y)$, where E_x and E_y are the components of the electric field in two dimensions. Show that the static Maxwell equations in free space imply that $E^*(z)$ is an analytic function of z.
6. Show that if we write $z = re^{i\theta}$, the Cauchy-Riemann equations become, in terms of r and θ,

$$\frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta}, \qquad \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}.$$

7. Analyze the function $z^{1/3}$ in terms of the Riemann sheet concepts discussed in Section 6.2.
8. Using the results of Problem 6.6, show that $\log z$, $z^{1/2}$, and $z^{1/3}$ are analytic everywhere on their appropriate Riemann sheets except at $z = 0$.
9. Show that the two-sheeted Riemann surface for $w(z) = \sqrt{(z - a)(z - b)}$ can be cut and joined along the line segment $[a, b]$ of the real axis (assume for convenience that a and b are real). Make an appropriate definition of $w(z)$ in terms of the polar representation of complex numbers, including a definition of the range of the phases.
10. Find the Riemann surface on which $\sqrt{z - a} + \sqrt{z - b}$ (a, b real and positive) is a single-valued function, analytic except at $z = a$ and $z = b$.
11. Find the Riemann surface on which $\sqrt{(z - 1)(z - 2)(z - 3)}$ is a single-valued function, analytic except at $z = 1, 2, 3$.
12. Show that $w(z) = z^\alpha$ $(0 < \alpha < 1)$, where α is irrational, can be made single-valued only on an infinite-sheeted Riemann surface. Discuss how the sheets should be connected.
13. Show that, according to the definitions of Section 6.2,

$$\tan(\tan^{-1} z) = z, \qquad \log(e^z) = z + 2\pi n i.$$

14. *The mean-value theorem.* Prove that for charge-free two-dimensional space the value of the electrostatic potential at any point is equal to the average of the potential over the surface of *any* circle centered on that point. Do this by considering the electrostatic potential as the real part of an analytic function.
15. Find all the singularities of $\tanh z$.
16. Explain why one of the following definite integrals is meaningful and the other meaningless. Evaluate the one that makes sense, writing the answer in the form $a + bi$.

$$\text{a)} \int_{-1}^{1} z^* \, dz, \qquad \text{b)} \int_0^i \sin 2z \, dz.$$

17. Compute, using Cauchy's integral formula,

a) $\displaystyle\oint_C \frac{e^z\,dz}{z - \pi i/2}$, $\quad C =$ boundary of a square with sides $x = \pm 2$, $y = \pm 2$.

b) $\displaystyle\oint_C \frac{dz}{z^2 + 2}$, $\quad C =$ circle of radius 1, center i.

(Both contours are oriented counterclockwise.)

18. Show that

a) $\displaystyle\oint_C \frac{z^4 + 2z + 1}{(z - z_0)^4}\,dz = 8\pi i z_0$, where C is any closed contour containing z_0.

b) $\displaystyle\oint_C \frac{\cosh z}{z^{n+1}}\,dz = \frac{2\pi i}{n!}\,\frac{[1 + (-1)^n]}{2}$, where C is any closed contour containing the origin.

19. *Cauchy's integral formula for a polynomial.* Let $P(z)$ be a polynomial. Using Cauchy's Theorem, prove that

$$\frac{1}{2\pi i}\oint_C \frac{P(z)}{z - a}\,dz = P(a)\,,$$

where C encloses the point a. [*Hint*: Consider

$$Q(z) = \frac{P(z) - P(a)}{z - a},$$

which is also a polynomial (why?), and evaluate the integral you want from this expression.]

20. Prove that if $f(z)$ is analytic within and on a closed contour C, and α and β are two distinct points within C, then

$$\frac{1}{2\pi i}\oint_C \frac{f(z)}{(z - \alpha)(z - \beta)}\,dz = \frac{f(\alpha)}{\alpha - \beta} + \frac{f(\beta)}{\beta - \alpha}.$$

From this result, deduce Liouville's theorem, which states that a function must be a constant. [*Hint*: If for arbitrary α and β, $f(\alpha) = f(\beta)$, then $f(z) =$ const.]

21. *Morera's theorem.* Morera's theorem is a kind of converse to the Cauchy-Goursat theorem. It states that if $f(z)$ is continuous in a region R and if $\oint f(z)\,dz = 0$ for any C inside R, then $f(z)$ is analytic inside R. Prove this theorem. Morera's theorem gives us another way of establishing the analyticity of a function. We can use it instead of checking the Cauchy-Riemann conditions *and* the sometimes troublesome additional requirement for analyticity, the continuity of the derivative of $f(z) = u + iv$, that is, continuity of the four first partial derivatives of u and v.

22. *Conformal mappings.* Consider the analytic function $w = f(z)$ as transforming the complex z-plane into a complex w-plane. Suppose that two curves in the z-plane, F_z^1 and F_z^2, meet at an angle α at the point z_0 in the z-plane. Prove that the transformed curves in the w-plane, F_w^1 and F_w^2, meet at the same angle in the w-plane if $f'(z_0) \neq 0$. The requirement that $f'(z_0) \neq 0$ is essential for conformality at z_0. Consider the analytic function $w = z^2$, for which $w'(0) = 0$. The coordinate axes themselves, which pass through the origin and are separated by an angle of 90°, are mapped into lines that are separated by 180°. The angle between any two straight lines through the origin in the z-plane will be doubled in the w-plane. Thus the mapping is not conformal at the origin. However, it is conformal everywhere else.

23. *Poisson's formula*

a) Let $f(z)$ be an analytic function within and on the circle C of radius a. Prove that

$$f(z') = \frac{1}{2\pi i}\oint_C \left[\frac{f(z)}{z - z'} - \frac{f(z)}{z - (a^2/z'^*)}\right] dz\,,$$

if $z' = re^{i\phi}$, $r < a$, in polar coordinates located at the center of C.

b) From the above formula, derive Poisson's formula:

$$f(re^{i\phi}) = \frac{1}{2\pi}\int_0^{2\pi} \frac{a^2 - r^2}{a^2 + r^2 - 2ar\cos(\theta - \phi)} f(ae^{i\theta})\, d\theta\,.$$

c) From Poisson's formula, prove that if a function is analytic throughout and on a circle, then the value of the function at the center of the circle, $f(0)$, is the average of its boundary values on the circle (mean value theorem).

d) Now, starting with the equation

$$f(z') = \frac{1}{2\pi i}\oint_C \left[\frac{f(z)}{z - z'} + \frac{f(z)}{z - (a^2/z'^*)}\right] dz\,,$$

[which differs from the equation of part (a) by the + sign, but holds for the same reason], derive [using part (c) eventually]

$$f(z') = f(0) - \frac{iar}{\pi}\int_0^{2\pi} \frac{\sin(\theta - \phi) f(ae^{i\theta})}{a^2 + r^2 - 2ar\cos(\theta - \phi)}\, d\theta\,.$$

e) Letting $f(z') = f(re^{i\phi}) = u(r, \phi) + iv(r, \phi)$, etc., deduce from (d) the formulas

$$u(a, \phi) = u(0) + \frac{1}{2\pi} P\int_0^{2\pi} v(a, \theta) \cot\left(\frac{\theta - \phi}{2}\right) d\theta\,,$$

$$v(a, \phi) = v(0) - \frac{1}{2\pi} P\int_0^{2\pi} u(a, \theta) \cot\left(\frac{\theta - \phi}{2}\right) d\theta\,.$$

These formulas express the real part of an analytic function on a circle in terms of its imaginary part on the circle, and vice versa. Many further applications of these formulas are possible.

24. *An alternative approach to Hilbert transforms*

a) Show that if $f(z)$ is analytic in the upper half z-plane, then if $a = \alpha + i\beta$ $(\beta > 0)$,

$$\frac{1}{2\pi i}\oint_C \left[\frac{f(z)}{z - a} + \frac{f(z)}{z - a^*}\right] dz = f(a)\,,$$

where C is a semicircle of arbitrarily large radius with the real axis for its base.

b) If $|f(z)| \to 0$ as $|z| \to \infty$ in the upper half-plane, then show that

$$\frac{1}{2\pi i}\int_{-\infty}^{\infty} \left[\frac{f(x)}{x - a} + \frac{f(x)}{x - a^*}\right] dx = f(a)\,.$$

Combine fractions to obtain

$$\frac{1}{i\pi}\int_{-\infty}^{\infty} \frac{(x - \alpha) f(x)}{(x - \alpha)^2 + \beta^2}\, dx = f(a)\,.$$

c) If $f(z) = u(x, y) + iv(x, y)$, show that (b) reduces to

$$u(\alpha, \beta) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{(x - \alpha) v(x, 0)}{(x - \alpha)^2 + \beta^2}\, dx\,, \quad v(\alpha, \beta) = -\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{(x - \alpha) u(x, 0)}{(x - \alpha)^2 + \beta^2}\, dx\,.$$

d) Hence show that if we let $\beta = 0$,

$$u(\alpha, 0) = \frac{1}{\pi} P\int_{-\infty}^{\infty} \frac{v(x, 0)}{x - \alpha}\, dx\,, \qquad v(\alpha, 0) = -\frac{1}{\pi} P\int_{-\infty}^{\infty} \frac{u(x, 0)}{x - \alpha}\, dx\,,$$

which is, apart from notation, the Hilbert transform pair derived in Section 6.5 [Eqs. (6.19a) and (6.19b)].

25. Consider the integral

$$I_{\pm} = \lim_{\epsilon \to 0} \int_{-A}^{+B} \frac{f(x)}{x \pm i\epsilon}\, dx\,,$$

where A and B are positive real numbers and $\epsilon > 0$.

a) Show that $I_{\pm}$ can be written as

$$I_{\pm} = \lim_{\epsilon \to 0} \int_{-A}^{-\delta} \frac{xf(x)}{x^2 + \epsilon^2}\, dx + \lim_{\epsilon \to 0} \int_{+\delta}^{+B} \frac{xf(x)}{x^2 + \epsilon^2}\, dx + \lim_{\epsilon \to 0} \int_{-\delta}^{+\delta} \frac{xf(x)}{x^2 + \epsilon^2}\, dx$$
$$\mp i \lim_{\epsilon \to 0} \epsilon \int_{-A}^{B} \frac{f(x)}{x^2 + \epsilon^2}\, dx\,,$$

where δ is a small positive number.

b) Use plausible arguments (of the type found, for example, in Section 5.3) to show that $I_{\pm}$ can be written as

$$I_{\pm} = \int_{-A}^{-\delta} \frac{f(x)}{x}\, dx + \int_{\delta}^{+B} \frac{f(x)}{x}\, dx + f(0) \lim_{\epsilon \to 0} \int_{-\delta}^{\delta} \frac{x\, dx}{x^2 + \epsilon^2} \mp i f(0) \lim_{\epsilon \to 0} \epsilon \int_{-A}^{B} \frac{dx}{x^2 + \epsilon^2}\,.$$

c) Show that the third term of the equation in (b) vanishes by symmetry. Doing the fourth integral explicitly, show that

$$I_{\pm} = P\int_{-A}^{B} \frac{f(x)}{x}\, dx \mp i f(0) \lim_{\epsilon \to 0} \left[\tan^{-1}\left(\frac{B}{\epsilon}\right) + \tan^{-1}\left(\frac{A}{\epsilon}\right) \right].$$

d) Let $\epsilon \to 0$ and use the definition of the δ-function to write, finally,

$$\lim_{\epsilon \to 0} \int_{-A}^{+B} \frac{f(x)}{x \pm i\epsilon}\, dx = P\int_{-A}^{+B} \frac{f(x)}{x}\, dx \mp i\pi f(0) = P\int_{-A}^{B} \frac{f(x)}{x}\, dx \mp i\pi \int_{-A}^{B} \delta(x) f(x)\, dx\,.$$

This is often written, rather cryptically, as the symbolic formula

$$\lim_{\epsilon \to 0} \frac{1}{x \pm i\epsilon} = P\frac{1}{x} \mp i\pi\delta(x)\,.$$

26. Solve the integral equation

$$\frac{1}{\pi} P\int_{-\infty}^{\infty} \frac{u(x)}{x - t}\, dx = \frac{1}{1 + t^2} \equiv -v(t)$$

by finding the Hilbert transform of $v(t)$. Find a complex function $f(z) = u(x, y) + iv(x, y)$ such that $u(x, 0) \equiv u(x) = Re(f(z))$ and $v(x, 0) \equiv v(x) = Im(f(z))$ and verify that this complex function has the two properties (what are they?) that ensure that $u(x)$ and $v(x)$ are a Hilbert transform pair.

27. Derive a dispersion relation for $f(z)$ when $|f(z)/z|$ tends to a constant as $|z| \to \infty$.

28. Prove that a uniformly convergent series of analytic functions can be integrated term by term. Hence show that a uniformly convergent series of analytic functions is an analytic function.

[*Hint*: Use Morera's theorem of Problem 21.]

29. Find the order of the pole at the origin, the residue there, and the integral around a (small) path C enclosing the origin, but no other singularities, for each of the following functions:

a) $\cot z$, b) $\csc^2 z \log(1-z)$, c) $\dfrac{z}{\sin z - \tan z}$.

30. What are the positions and natures of the singularities and the residues at these singularities of the following functions in the z-plane, excluding the point at infinity?

a) $f_1(z) = \dfrac{\cot \pi z}{(z-1)^2}$, b) $f_2(z) = \dfrac{1}{z(e^z - 1)}$.

[*Note*: Both these functions have nonisolated and hence essential singularities at infinity because the origin for $f_1(1/z)$ and $f_2(1/z)$ is a limit point of poles.]

31. Problems on the evaluation of real definite integrals by contour integration:

a) $\displaystyle\int_0^{2\pi} \frac{d\theta}{a + b\sin\theta} = \frac{2\pi}{(a^2 - b^2)^{1/2}}, \qquad a > b > 0.$

b) $\displaystyle\int_0^{2\pi} \frac{\sin^2\theta\, d\theta}{a + b\cos\theta} = \frac{2\pi}{b^2}[a - (a^2 - b^2)^{1/2}], \qquad a > b > 0.$

c) $\displaystyle\int_0^{2\pi} \frac{d\theta}{(a + b\cos\theta)^2} = \frac{2\pi a}{(a^2 - b^2)^{3/2}}, \qquad a > b > 0.$

d) $\displaystyle\int_0^{\infty} \frac{dx}{1 + x^4} = \frac{\pi}{2\sqrt{2}}.$ e) $\displaystyle\int_0^{\infty} \frac{x^2\, dx}{(a^2 + x^2)^3} = \frac{\pi}{16a^3}.$

f) $\displaystyle\int_0^{\infty} \frac{\sin x\, dx}{x(a^2 + x^2)} = \frac{\pi}{2a^2}(1 - e^{-a}).$ g) $\displaystyle\int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2}\, dx = \pi.$

h) $\displaystyle\int_0^{\infty} \frac{x^{2a-1}}{b^2 + x^2}\, dx = \frac{\pi b^{2(a-1)}}{2} \csc \pi a, \qquad 0 < a < 1.$

i) $\displaystyle\int_0^{\infty} \frac{\log x}{b^2 + x^2}\, dx = \frac{\pi \log b}{2b}.$

j) $\displaystyle P\int_{-\infty}^{\infty} \frac{1}{(\omega' - \omega_0)^2 + a^2} \frac{1}{\omega' - \omega}\, d\omega' = \frac{\pi}{a} \frac{\omega_0 - \omega}{(\omega_0 - \omega)^2 + a^2}.$

k) $\displaystyle\int_0^{\infty} \frac{dx}{x^3 + a^3} = \frac{2\pi}{3a^2\sqrt{3}}.$

l) Consider the real integral

$$G(x, x', \tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{ik(x-x')} e^{-k^2\tau}\, dk,$$

where τ is real. In this case it is easy to show that

$$G(x, x', \tau) = \frac{1}{(4\pi\tau)^{1/2}} e^{-(x-x')^2/4\tau}.$$

Prove that this result still holds when τ is pure imaginary. [*Hint*: The value of the integral

$$\int_0^{\infty} e^{-iu^2}\, du$$

may be found by considering the integral of e^{-z^2} around the boundary of the circular sector $0 \le \theta \le \pi/4$, $0 \le r \le R$. In the limit $R \to \infty$, *prove* that the contribution to the integral over the circular arc goes to zero. The contributions over the remaining straight segments of the contour provide the value of the integral needed in the problem.]

Note: This contour integral provides the values of the two real definite integrals:

$$\int_{-\infty}^{\infty} \sin(x^2)\,dx = \int_{-\infty}^{\infty} \cos(x^2)\,dx = (\pi/2)^{1/2},$$

by taking real and imaginary parts. By making the change of variable $x^2 = t$, this result can be transformed into

$$\int_0^{\infty} \frac{\sin t}{\sqrt{t}}\,dt = \int_0^{\infty} \frac{\cos t}{\sqrt{t}}\,dt = (\pi/2)^{1/2}.$$

32. We have seen how the function $\pi \cot \pi z$ may be used to sum certain series. But it is useless if the series is an alternating one. Show how alternating series may be summed by using the function $\pi \csc(\pi z)$ in place of $\pi \cot(\pi z)$. Use the general result to prove that

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^2} = \pi^2/12.$$

33. Consider the problem of evaluating the integral of $z^{1/2}$ from A to D along the circular path C, shown in Fig. 6.24. We take the cut to lie along the ray $\theta = \alpha$ and define $z^{1/2} \equiv r^{1/2}e^{i\theta/2}$ for $\alpha \leq \theta < \alpha + 2\pi$. Compare this with the value obtained by integrating $z^{1/2}$ from A to D along the "keyhole" path $ABCD$. Do the results agree? Should they agree? Why?

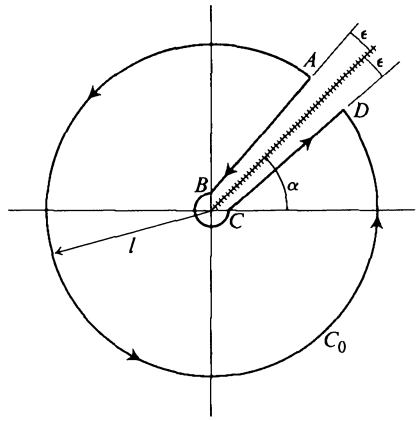

Fig. 6.24 The contours for Problem 6-34. The angles, ϵ, for the "keyhole" path are arbitrarily small. The radius of the circle, C_0, is l, and the cut is along the ray $\theta = \alpha$.

34. Consider the following response function, $G(t)$, which vanishes for negative values of the argument and which, for positive values, is given by

$$G(t) = G_0 \frac{\sin^2 \mu t}{t^{3/2}}.$$

Note that this function satisfies conditions (a), (b), (c) at the beginning of Section

6.6. Show that the Fourier transform, $g(\omega)$, is given by

$$g(\omega) = \frac{i-1}{2} G_0[\sqrt{\omega} - \tfrac{1}{2}\sqrt{\omega + 2\mu} - \tfrac{1}{2}\sqrt{\omega - 2\mu}] , \qquad \omega \geq 2\mu ;$$

$$= \frac{i-1}{2} G_0[\sqrt{\omega} - \tfrac{1}{2}\sqrt{\omega + 2\mu}] + \frac{i+1}{2} G_0[\tfrac{1}{2}\sqrt{2\mu - \omega}] , \qquad 0 \leq \omega \leq 2\mu ,$$

$$g(-\omega) = g^*(\omega) .$$

Show that the complex-valued function of z which reduces to the above result on the real axis is

$$g(z) = \frac{i-1}{2} G_0[\sqrt{z} - \tfrac{1}{2}\sqrt{z + 2\mu} - \tfrac{1}{2}\sqrt{z - 2\mu}] .$$

Thus we see that $g(z)$ has branch points on the real axis,a possibility which is *not* excluded by conditions (a), (b), and (c). Note that we can choose the cuts to lie in the lower half-plane, so $g(z)$ is analytic in the upper half-plane.

[*Hint*: the integrals mentioned in the note following Problem 6.31 will be of use.]

35. The θ-function (or step function) is defined as

$$\theta(x) = \begin{cases} 0 & \text{for } x < 0 , \\ \tfrac{1}{2} & \text{for } x = 0 , \\ 1 & \text{for } x > 0 . \end{cases}$$

Prove

a) $\theta(x) = \dfrac{1}{2} + \dfrac{1}{2\pi i} P\displaystyle\int_{-\infty}^{\infty} \frac{e^{i\omega x}}{\omega} d\omega ,$ b) $\dfrac{d\theta(x)}{dx} = \delta(x) .$

[*Hint*: See Eq. (5.63)]

36. The Laguerre polynomials are generated by the generating function

$$\psi(x, t) = \frac{e^{-xt/(1-t)}}{1-t} = \sum_{n=0}^{\infty} \frac{L_n(x)}{n!} t^n , \qquad 0 < t < 1 .$$

They are given directly by the formula

$$L_n(x) = e^x \frac{d^n}{dx^n}(x^n e^{-x}) , \qquad 0 \leqq x \leqq \infty ,$$

analogous to Rodrigues' formula for Legendre polynomials.

a) By differentiating the generating function with respect to x [getting $(1-t)\psi' = -t\psi$], derive the recursion relation

$$L_n' - nL_{n-1}' = -nL_{n-1} .$$

By differentiating $\psi(x, t)$ with respect to t, another recursion relation can be derived. Show that

$$L_{n+1} - (2n + 1 - x)L_n + n^2 L_{n-1} = 0 .$$

b) From these two recursion relations, derive Laguerre's differential equation

$$xL_n'' + (1 - x)L_n' + nL_n = 0 .$$

c) Show that $L_n(0) = n!$

d) Derive the generating function from Rodrigues' formula by contour integration.

37. In Section 5.10, we defined the Hermite polynomials in terms of the generating function

$$e^{-t^2+2tx} = \sum_{n=0}^{\infty} \frac{H_n(x)}{n!} t^n , \qquad t > 0 ,$$

and deduced from this the Rodrigues formula:

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2} .$$

Now prove the converse: that Rodrigues' formula implies the generating function. Enroute, establish the integral representation

$$H_n(x) = \frac{(-1)^n n!}{2\pi i} e^{x^2} \oint_C \frac{e^{-z^2}}{(z-x)^{n+1}} \, dz ,$$

where C encloses the point x.

38. Prove the integral representation for the Hermite polynomials:

$$H_n(x) = \frac{2^n}{\pi^{1/2}} \int_{-\infty}^{\infty} (x + it)^n e^{-t^2} \, dt .$$

Use this result to derive an explicit series for the Hermite polynomials:

$$H_n(x) = \sum_{\substack{r=0 \\ (2r \leq n)}}^{n} \frac{(-1)^r n!}{r!\,(n-2r)!} (2x)^{n-2r} .$$

39. Show that
    a) $\Gamma(x+1) = x\Gamma(x)$ for $x > 0$
    b) $\Gamma(n+1) = n!$, where n is a positive integer.

FURTHER READINGS

CHURCHILL, R. V., *Complex Variables and Applications*. Second edition. New York: McGraw-Hill, 1960. An excellent text of great clarity, it is ideal for self-study. Churchill's books have influenced a generation of physicists (including the authors).

FRANKLIN, P., *Functions of Complex Variables*. Englewood Cliffs, N.J.: Prentice Hall, 1958.

GOURSAT, E., *Functions of a Complex Variable*. New York: Dover Publications, 1966. Very readable.

HILLE, E., *Analytic Function Theory*, Volumes I and II. Boston: Ginn, 1959. An exhaustive account.

KNOPP, K., *Theory of Functions*, Parts I and II. New York: Dover, 1945. There are two accompanying problem books.

LANDAU, E., *Foundations of Analysis*. New York: Chelsea, 1960. An axiomatic treatment of the various number systems, including complex numbers.

SESHU, S., and N. BALABARIAN, *Linear Network Analysis*. New York: Wiley, 1959. This book treats engineering applications of dispersion relations.

SCHWEBER, S., *An Introduction to Relativistic Quantum Field Theory*. Evanston, Ill.: Row Peterson, 1961. See p. 411 for a discussion of causality and dispersion relations.

TITCHMARSH, E. C., *Introduction to the Theory of Fourier Integrals*. New York: Oxford University Press, 1937. Chapter 5 provides a rigorous treatment of the Hilbert transform pair.

WEYL, H., *Philosophy of Mathematics and Natural Science*. Princeton, N. J.: Princeton University Press, 1949. Chapter II is devoted to the various number systems.

CHAPTER 7

GREEN'S FUNCTIONS

INTRODUCTION

In this chapter, we turn our attention to integral operators. It turns out that there is a particularly close connection between differential and integral operators; and the boundary conditions with which one is familiar from the theory of differential equations play a crucial role in linking the two. The connection between them is essentially the theory of Green's functions, an area in theoretical physics which has become particularly important in connection with modern developments in quantum mechanics. We cannot hope to give a comprehensive treatment of this enormous subject in just one chapter, but we shall try to illustrate the main ideas. Green's functions can be useful in classical physics, but our emphasis here will be primarily on their role in quantum mechanics, for it has been largely through the formulation in terms of Green's functions that the mathematical structure of quantum mechanics, both in the relativistic and nonrelativistic regimes, has been clarified.

7.1 A NEW WAY TO SOLVE DIFFERENTIAL EQUATIONS

The simplest of all differential equations is undoubtedly

$$-i\frac{dy}{dx} = f(x) \ . \tag{7.1}$$

(We have put the i in front of the derivative for future convenience; note the analogy with the quantum-mechanical operator $p_x = -i\hbar d/dx$.) If we choose the initial condition to be $y(a) = y_0$, the solution to Eq. (7.1) may be written immediately:

$$y(x) = y_0 + i\int_a^x f(x')\,dx' \ . \tag{7.2}$$

Thus we have already solved a simple differential equation by integration.

But this is really a trivial example, because in these terms every integral one writes may be dignified by the title "A Solution to a Differential Equation." However, let us push the problem a bit further. Imagine that x lies within some closed interval $[a, b]$. Then we may rewrite Eq. (7.2) as

$$y(x) = y_0 + i\int_a^b \theta(x - x')f(x')\,dx' \ , \tag{7.3}$$

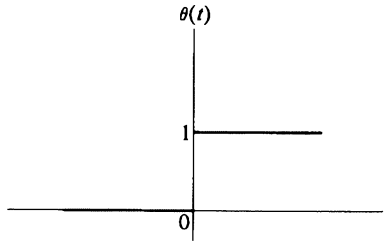

Fig. 7.1 The Heaviside function.

where $\theta(t)$ is the discontinuous step function defined by

$$\theta(t) = \begin{cases} 1\,, & t > 0\,, \\ 0\,, & t \leq 0\,, \end{cases} \tag{7.4}$$

$\theta(t)$ is sometimes called the Heaviside function (Fig. 7.1). Its presence in the integrand in Eq. (7.3) serves to cut off the integration over x' at $x = x'$; thus Eq. (7.3) is equivalent to Eq. (7.2). We note that $y(x)$ has the form

$$y(x) = y_0 + Kf(x)\,, \tag{7.5}$$

where K is an integral operator defined by

$$Kf(x) \equiv i\int_a^b \theta(x - x')f(x')\,dx'\,. \tag{7.6}$$

Here $i\theta(x - x')$ is the *kernel* of the integral operator K, and when the kernel comes from the solution of an equation involving a differential operator, it is often referred to as the *Green's function* of that differential operator *for the relevant boundary conditions*. Thus

$$G_1(x, x') = i\theta(x - x') \tag{7.3$'$}$$

is the Green's function belonging to id/dx for a system subject to the boundary condition $y(a) = y_0$.

A more interesting example is the equation

$$\frac{d^2y}{dx^2} = f(x)\,, \tag{7.7}$$

where $y(x)$ is subject to the one-point boundary conditions, $y(a) = y_0$, $y'(a) = \bar{y}_0$. Integrating Eq. (7.7) once, we obtain

$$\frac{dy}{dx} = \bar{y}_0 + \int_a^x f(x')\,dx'\,;$$

a second integration gives

$$y(x) = y_0 + \bar{y}_0(x - a) + \int_a^x dx'' \int_a^{x''} dx'\,f(x')\,.$$

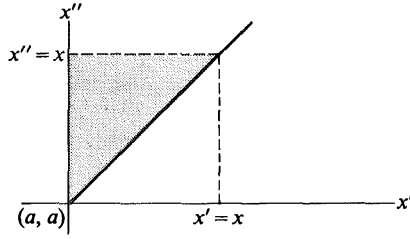

Fig. 7.2

In this expression we are integrating over a triangular region in the $x'x''$ plane (the shaded region in Fig. 7.2) by first integrating x' from $x' = a$ to $x' = x''$, and then integrating x'' from a to x. However, we can equally well integrate x'' from $x'' = x'$ to $x'' = x$ and then integrate x' from a to x. Since the function being integrated depends only on x', it is advantageous to change the order and limits of integration in this manner. We get

$$y(x) = y_0 + \bar{y}_0(x - a) + \int_a^x dx'\, f(x') \int_{x'}^x dx''$$

so that

$$y(x) = y_0 + \bar{y}_0(x - a) + \int_a^x (x - x')f(x')\, dx' . \tag{7.8}$$

The reader may check this by differentiating $y(x)$ twice and comparing the result with Eq. (7.3). Again, if we restrict x to the interval $[a, b]$, we can write

$$y(x) = y_0 + \bar{y}_0(x - a) + \int_a^b (x - x')\theta(x - x')f(x')\, dx' . \tag{7.9}$$

This solution for $y(x)$ has exactly the same form as the solution to Eq. (7.1). Using the terminology introduced earlier, we may say that the function

$$G_2(x, x') = (x - x')\theta(x - x') \tag{7.9'}$$

is the Green's function corresponding to the operator d^2/dx^2 for the boundary conditions $y(a) = y_0$, $y'(a) = \bar{y}_0$. Note that this Green's function is continuous while the Green's function of the operator d/dx is discontinuous (at $x = x'$).

In both Eqs. (7.3) and (7.9), the term on the right-hand side which is not under the integral is a solution of the homogeneous equation for the given boundary condition. We have

$$\frac{dy}{dx} = 0$$

in the case of Eq. (7.3), and

$$\frac{d^2y}{dx^2} = 0$$

in the case of Eq. (7.9). Therefore, the general solution to Eq. (7.7) is

$$y(x) = \alpha + \beta x + \int_a^b (x - x')\theta(x - x')f(x')\,dx' , \tag{7.10}$$

where α and β are to be determined by the boundary conditions.

For example, if we have a two-point boundary condition

$$y(0) = y_0,\ y(1) = y_1 ,$$

the solution interior to the interval [0, 1] will be

$$y(x) = \alpha + \beta x + \int_0^1 (x - x')\theta(x - x')f(x')\,dx' ,$$

where α and β are determined via the equations

$$y_0 = \alpha - \int_0^1 x'\theta(-x')f(x')\,dx' = \alpha ,$$

$$y_1 = \alpha + \beta + \int_0^1 (1 - x')\theta(1 - x')f(x')\,dx' .$$

Thus

$$\alpha = y_0 , \qquad \beta = y_1 - y_0 - \int_0^1 (1 - x')f(x')dx' .$$

Hence the solution looks like

$$y(x) = y_0 + (y_1 - y_0)x - \int_0^1 x(1 - x')f(x')\,dx' + \int_0^1 (x - x')\theta(x - x')f(x')\,dx' .$$

Calling

$$G_3(x, x') = -x(1 - x') + (x - x')\theta(x - x') ,$$

we have

$$y(x) = y_0 + (y_1 - y_0)x + \int_0^1 G_3(x, x')f(x')\,dx' . \tag{7.11}$$

Let us look at $G_3(x, x')$ more closely. When $0 \le x' \le x$,

$$G_3(x, x') = -x(1 - x') + (x - x') = -x'(1 - x) ;$$

when $x \le x' \le 1$,

$$G_3(x, x') = -x(1 - x') .$$

To summarize, the solution to Eq. (7.7) is given by Eq. (7.11), where $G_3(x, x')$ is

$$G_3(x, x') = \begin{cases} -x'(1 - x) , & 0 \le x' \le x , \\ -x(1 - x') , & x \le x' \le 1 . \end{cases} \tag{7.11'}$$

Using the function $\theta(x)$, $G_3(x, x')$ can also be written conveniently as

$$G_3(x, x') = xx' - x'\theta(x - x') - x\theta(x' - x) . \tag{7.11''}$$

Note that $G_3(x, x')$ vanishes at $x = 0$ and $x = 1$, and also at $x' = 0$ and $x' = 1$.

It is interesting to observe the dramatic effect which the boundary conditions have on the form of the Green's function. For one thing, the functional form changes. Moreover, the Green's function of Eq. (7.11″) is a symmetric function of x and x', since

$$G_3(x, x') = G_3(x', x);$$

but the Green's function of Eq. (7.9′) is not symmetric. However, both $G_2(x, x')$ and $G_3(x, x')$ are Green's functions of the operator d^2/dx^2; the only difference lies in the boundary conditions.

Let us now examine the effect of changing boundary conditions in a first-order equation. We return to Eq. (7.1) and this time try the boundary condition $y(1) = Cy(0)$. The solution will have the general form

$$y(x) = A + i\int_0^1 \theta(x - x')f(x')\,dx' .$$

Now we require that

$$y(1) = A + i\int_0^1 f(x')\,dx' = Cy(0) = CA;$$

here we have substituted both $x=1$ and $x=0$ into the above equation. From this we find immediately that

$$A = -\frac{i}{1 - C}\int_0^1 f(x')\,dx' .$$

Thus

$$y(x) = -\frac{i}{1 - C}\int_0^1 f(x')\,dx' + i\int_0^1 \theta(x - x')f(x')\,dx' ,$$

or

$$y(x) = \int_0^1 G_4(x, x')f(x')\,dx' , \tag{7.12}$$

where

$$G_4(x, x') = -i\left[\frac{1}{1 - C} - \theta(x-x')\right].$$

Thus

$$G_4(x, x') = \begin{cases} -\dfrac{iC}{1 - C}, & x > x' , \\ -\dfrac{i}{1 - C}, & x \leq x' . \end{cases}$$